

Singular Value Decomposition

Geometric Algorithms

Objectives

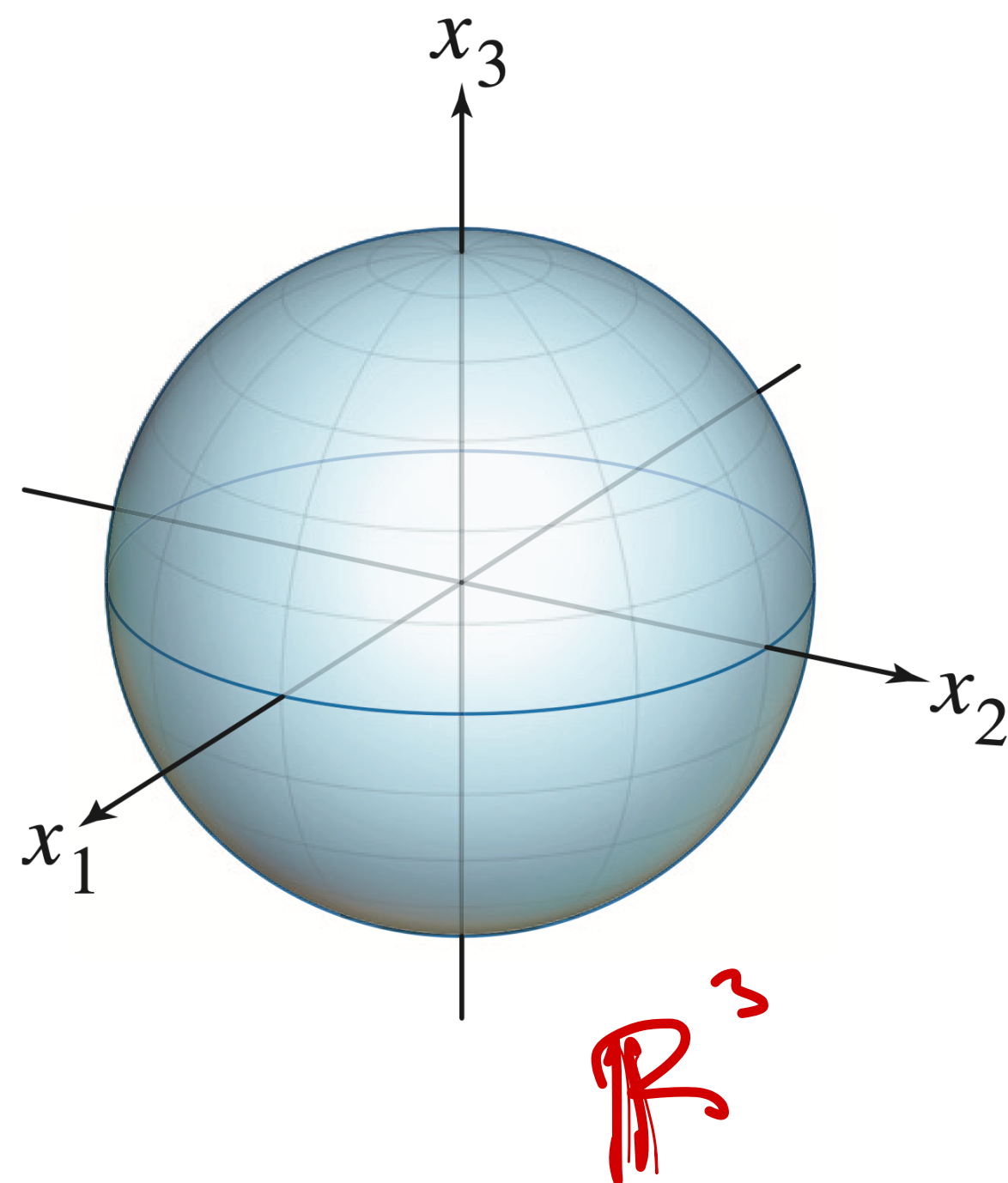
Introduce the **singular value decomposition**

(probably the most important decomposition for CS)

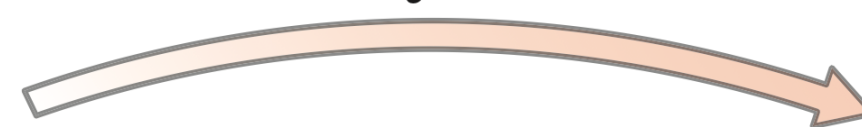
Motivation

Question

What shape is a the unit sphere after a linear transformation?



Multiplication
by A



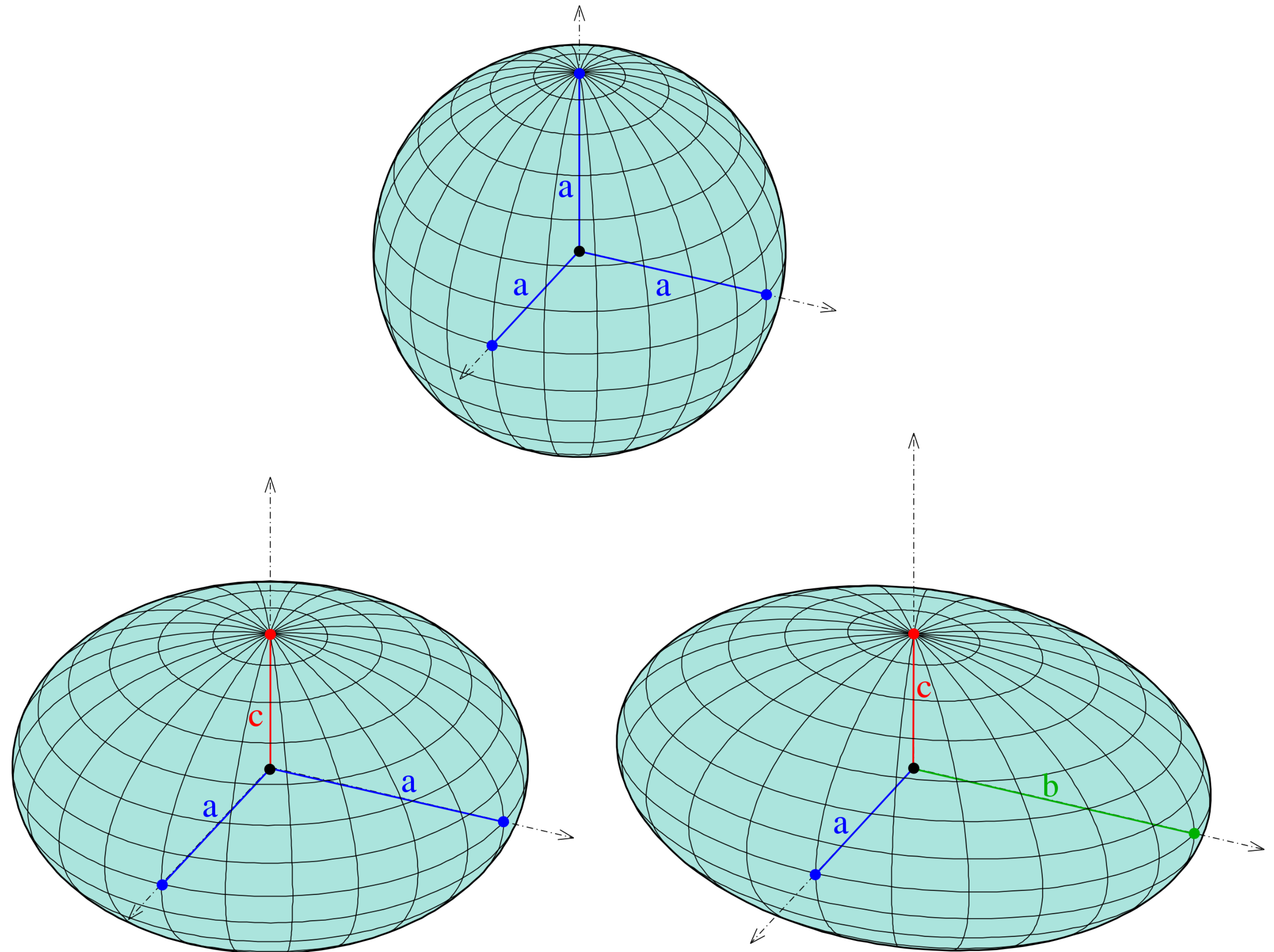
???

$\mathbb{R}^k$

Ellipsoids

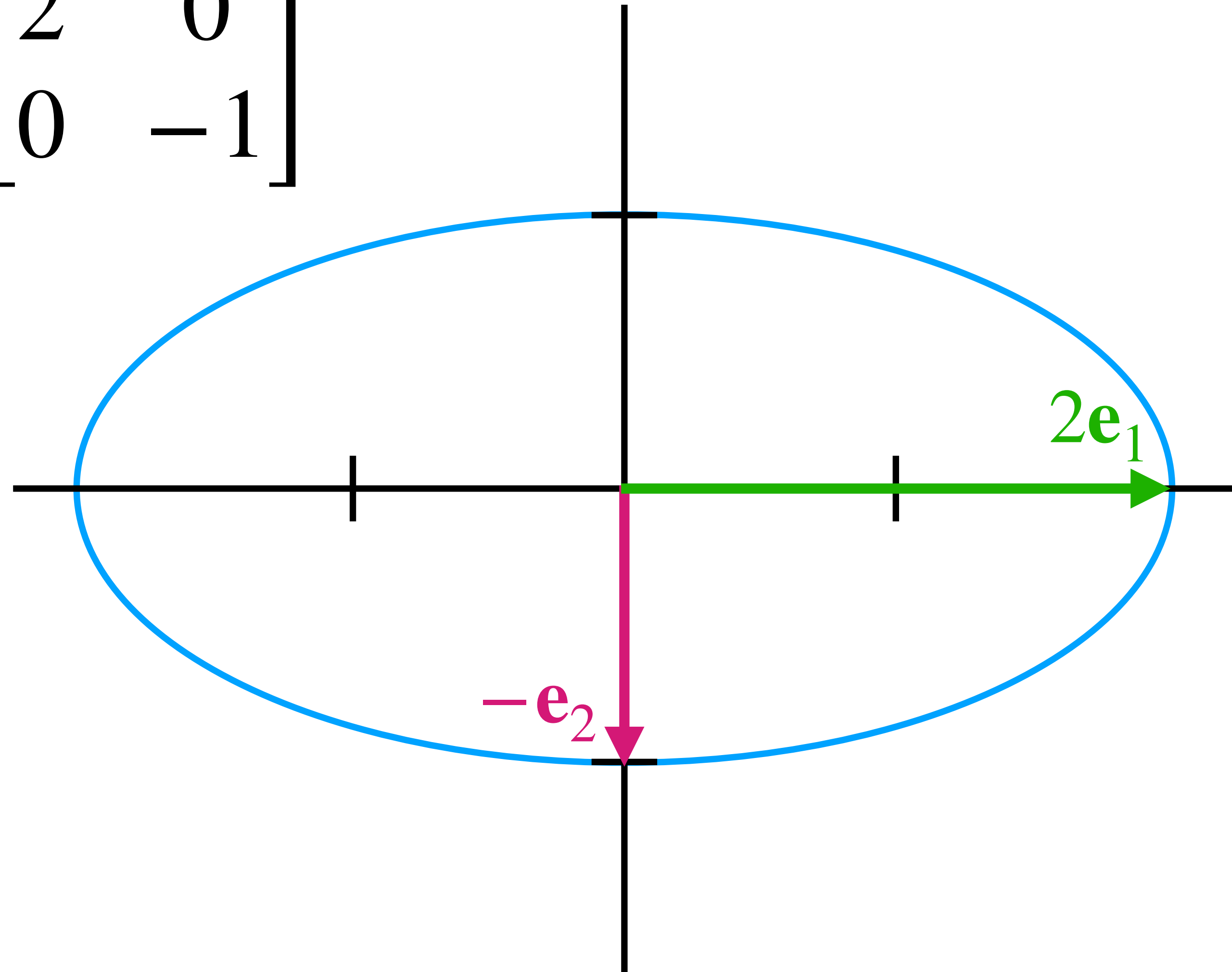
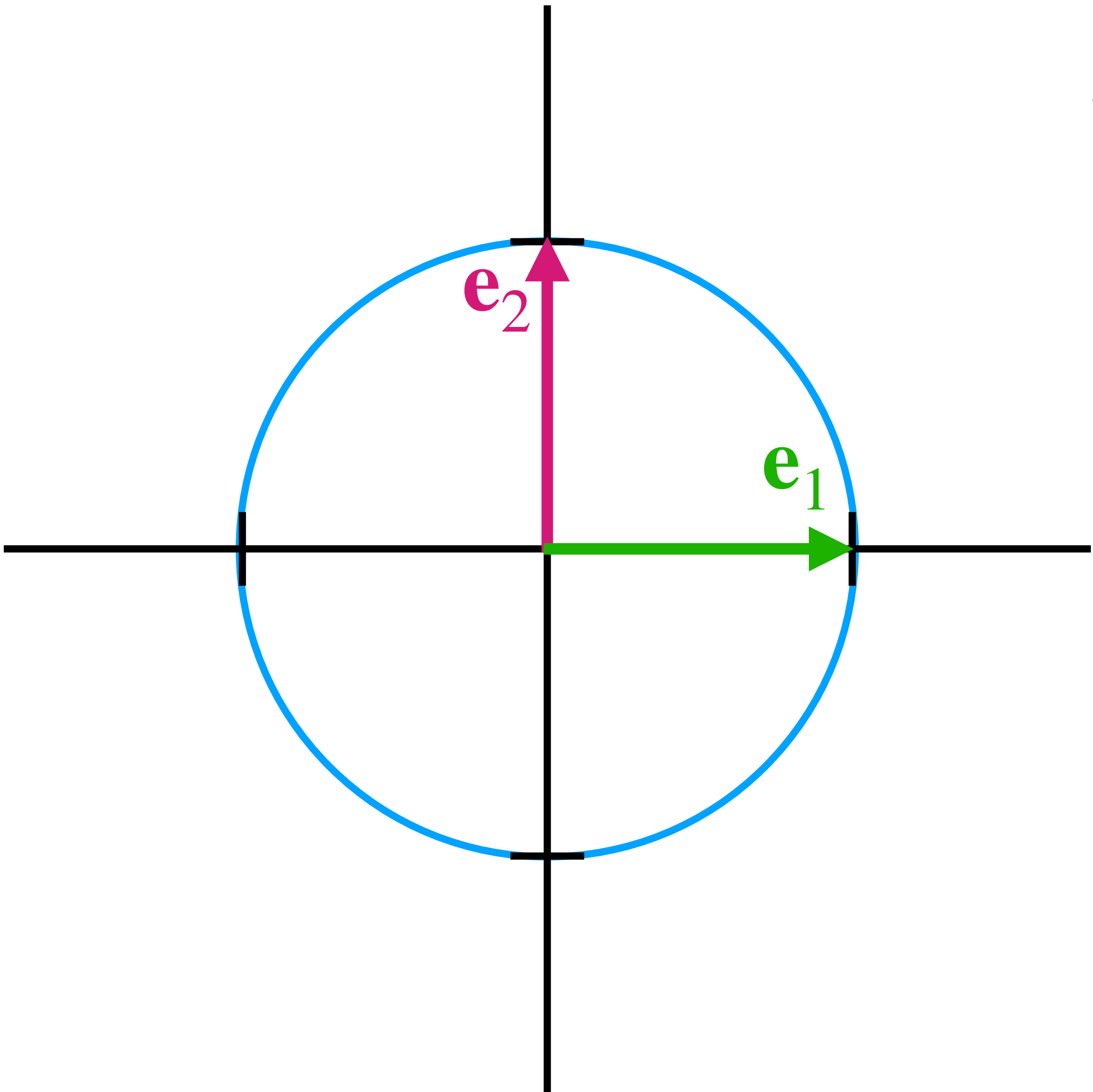
Ellipsoids are spheres
"stretched" in
orthogonal directions
called the
axes of symmetry or the
principle axes

Linear trans. map
spheres to ellipsoids

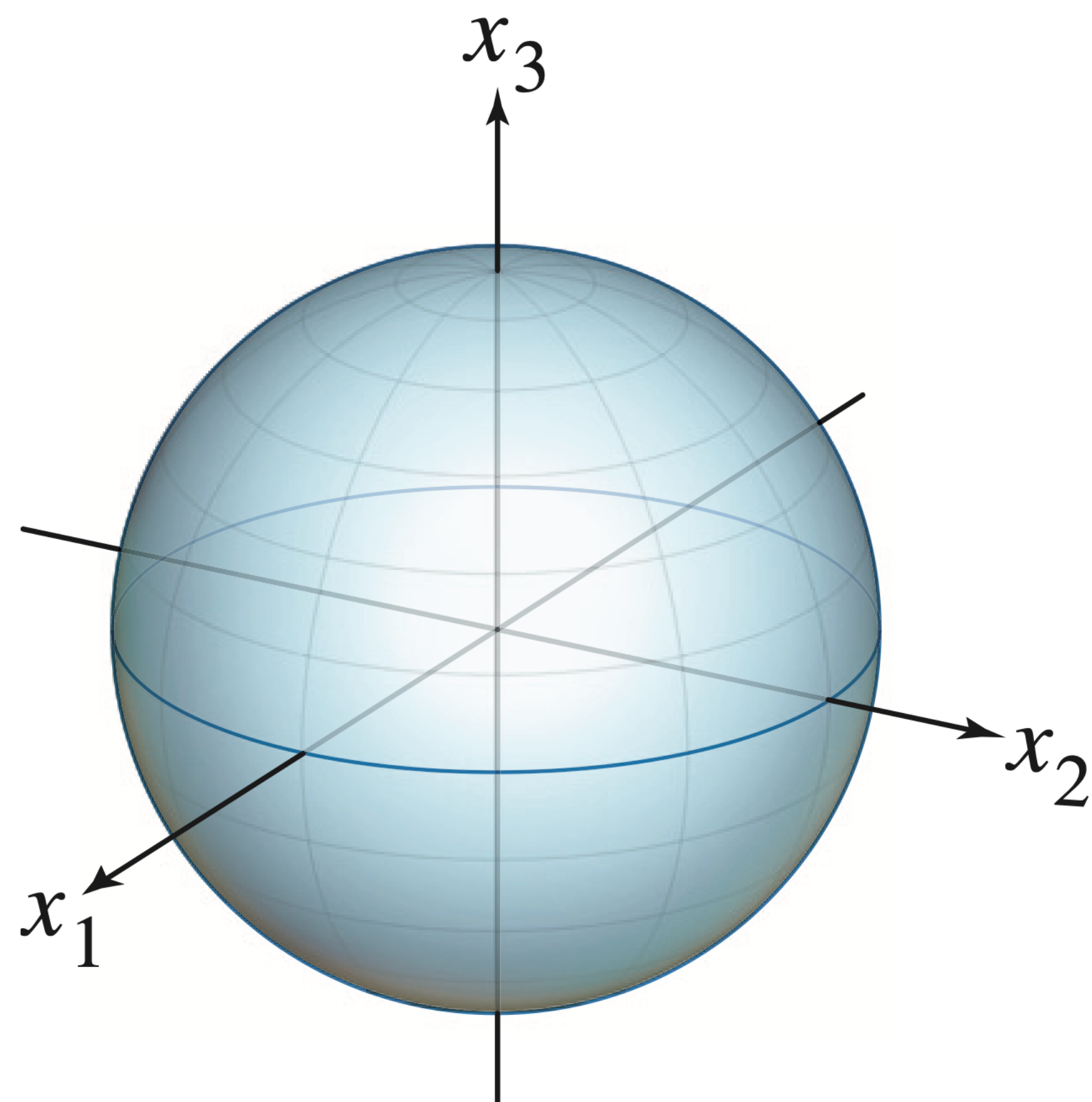


Simple Example : Scaling Matrices

$$A = \begin{bmatrix} 2 & 0 \\ 0 & -1 \end{bmatrix}$$

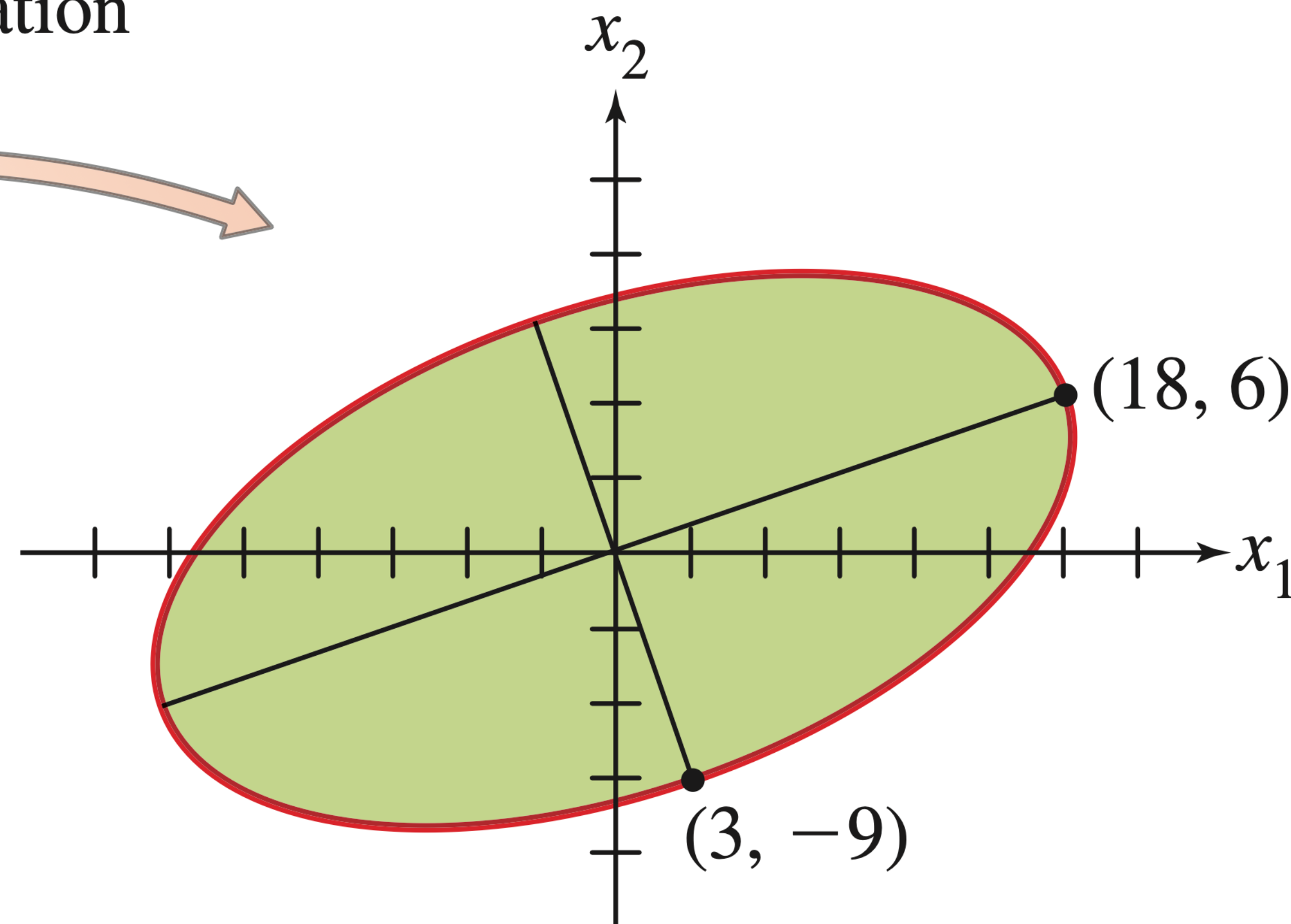
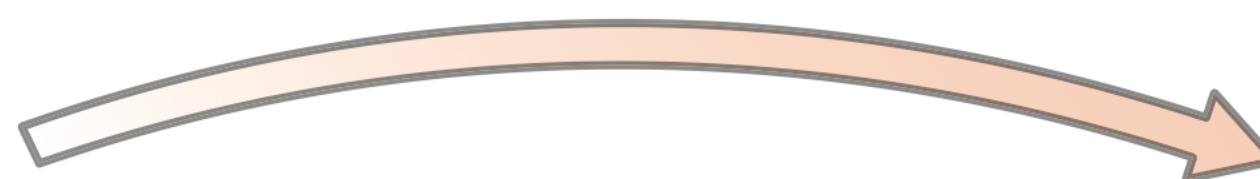


The Picture



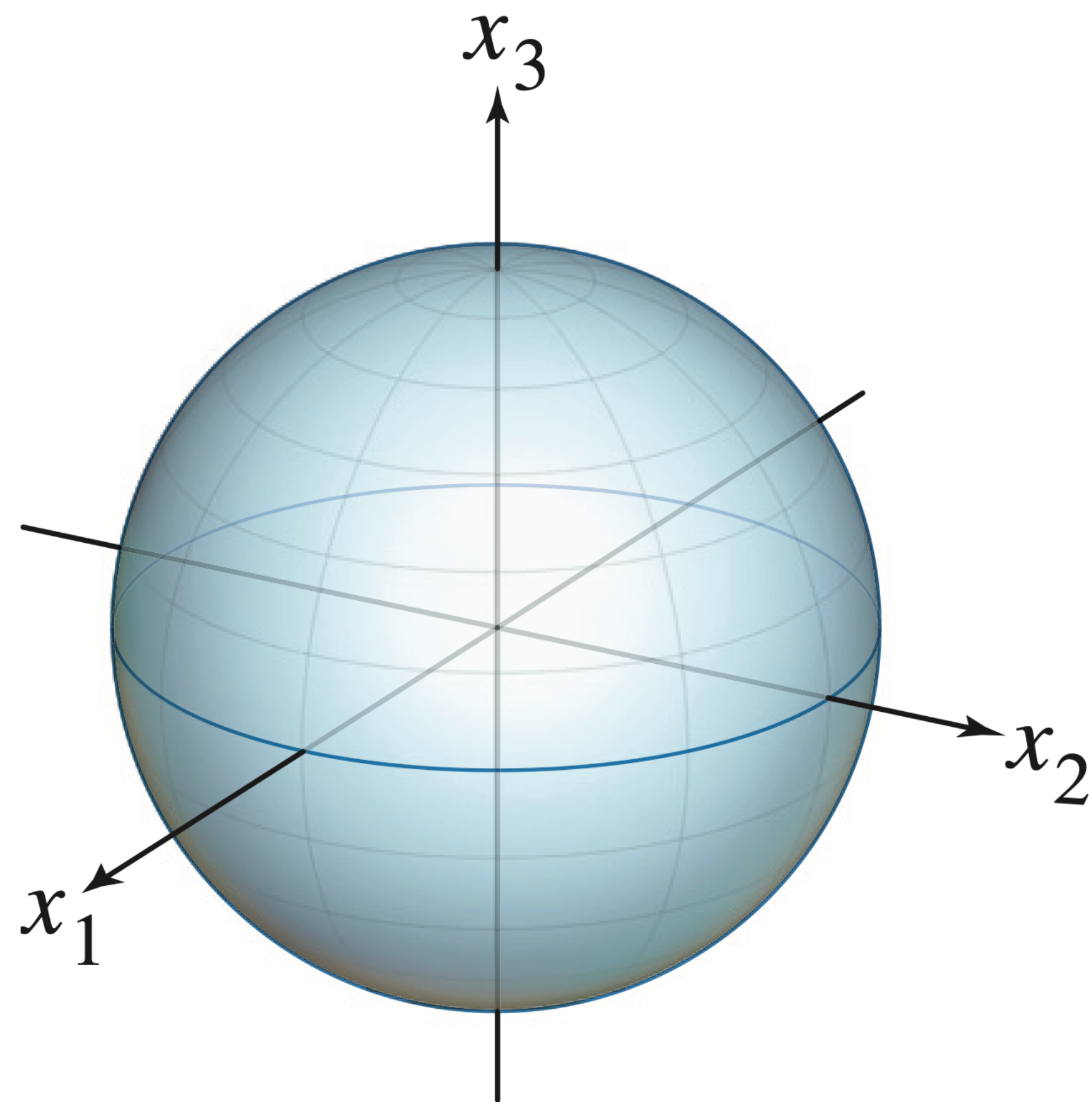
$$\mathbb{R}^{2 \times 3}$$

Multiplication
by A

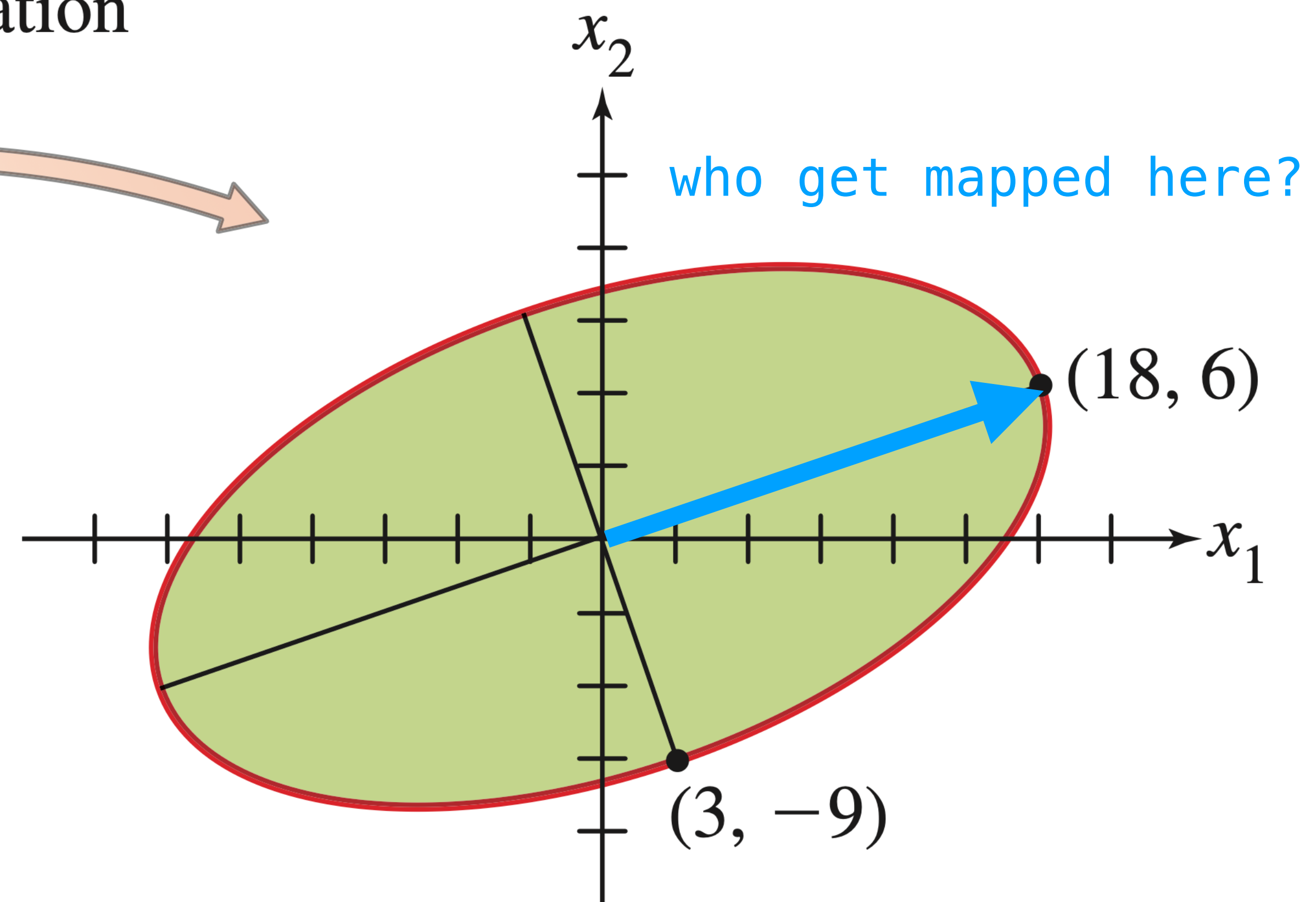
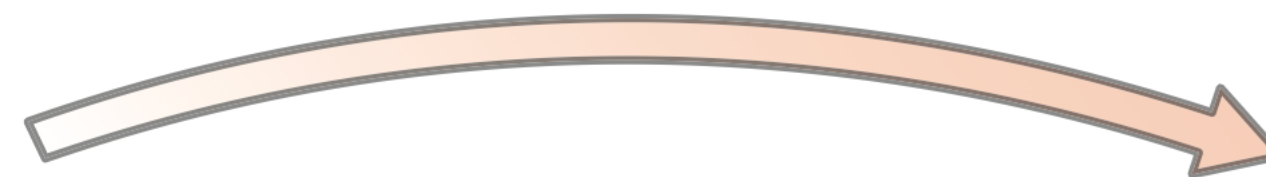


$$\underbrace{A}_{2 \times 3} \underbrace{\vec{r}}_{\mathbb{R}^3} = \underbrace{\vec{u}}_{\mathbb{R}^2}$$

The Picture

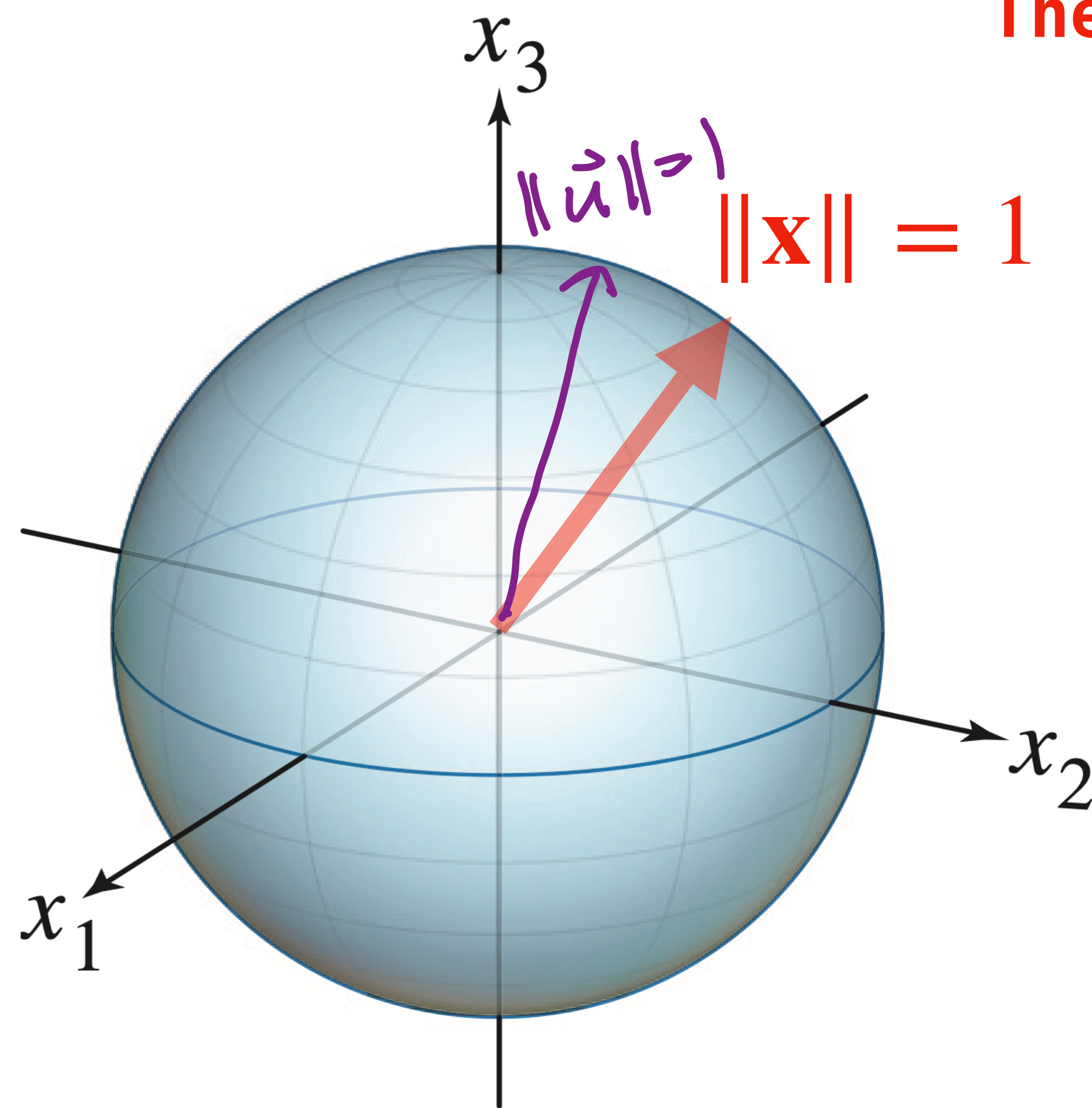


Multiplication
by A

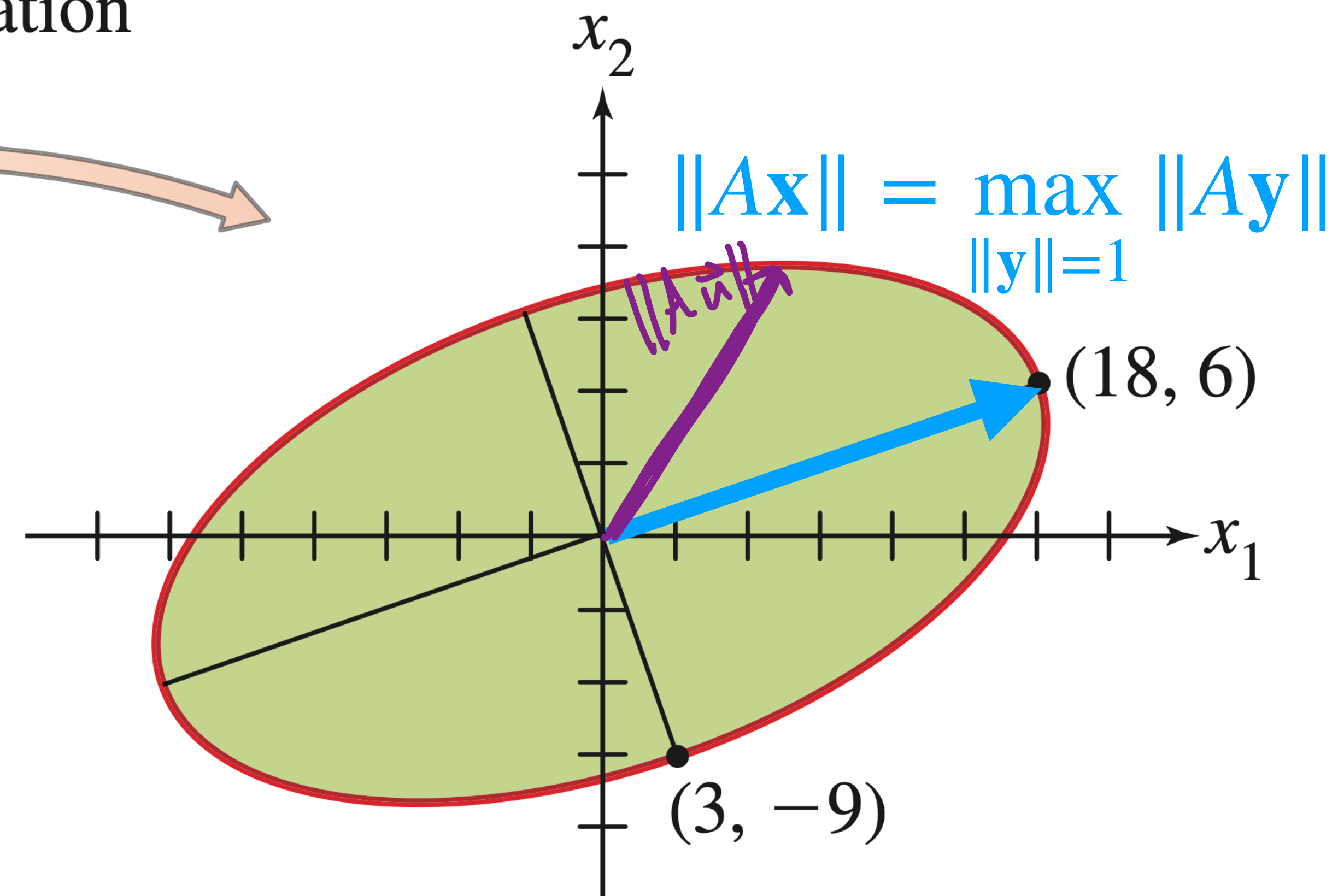


The Picture

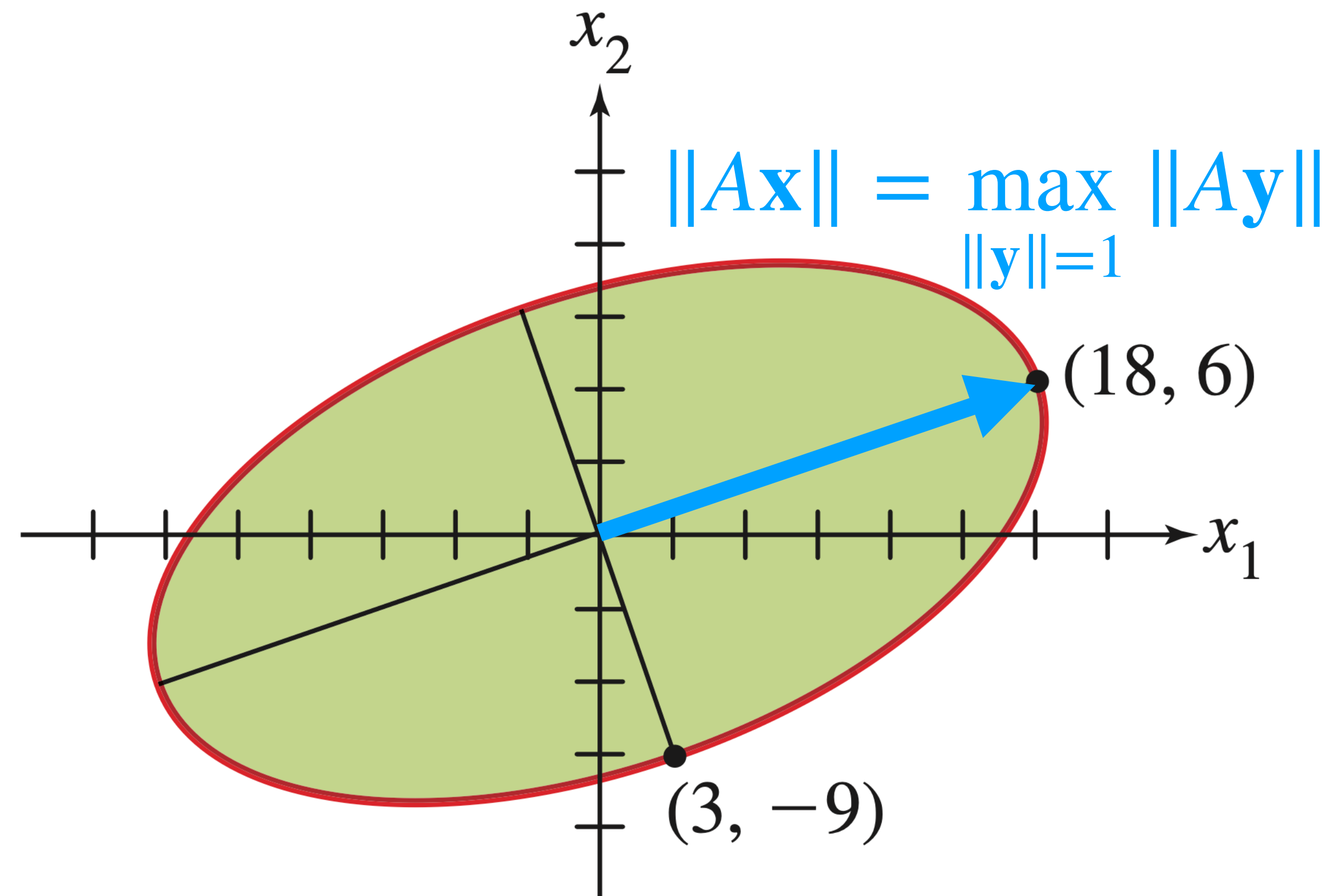
The longest end of the ellipse is the solution to a constrained optimization problem



Multiplication
by A

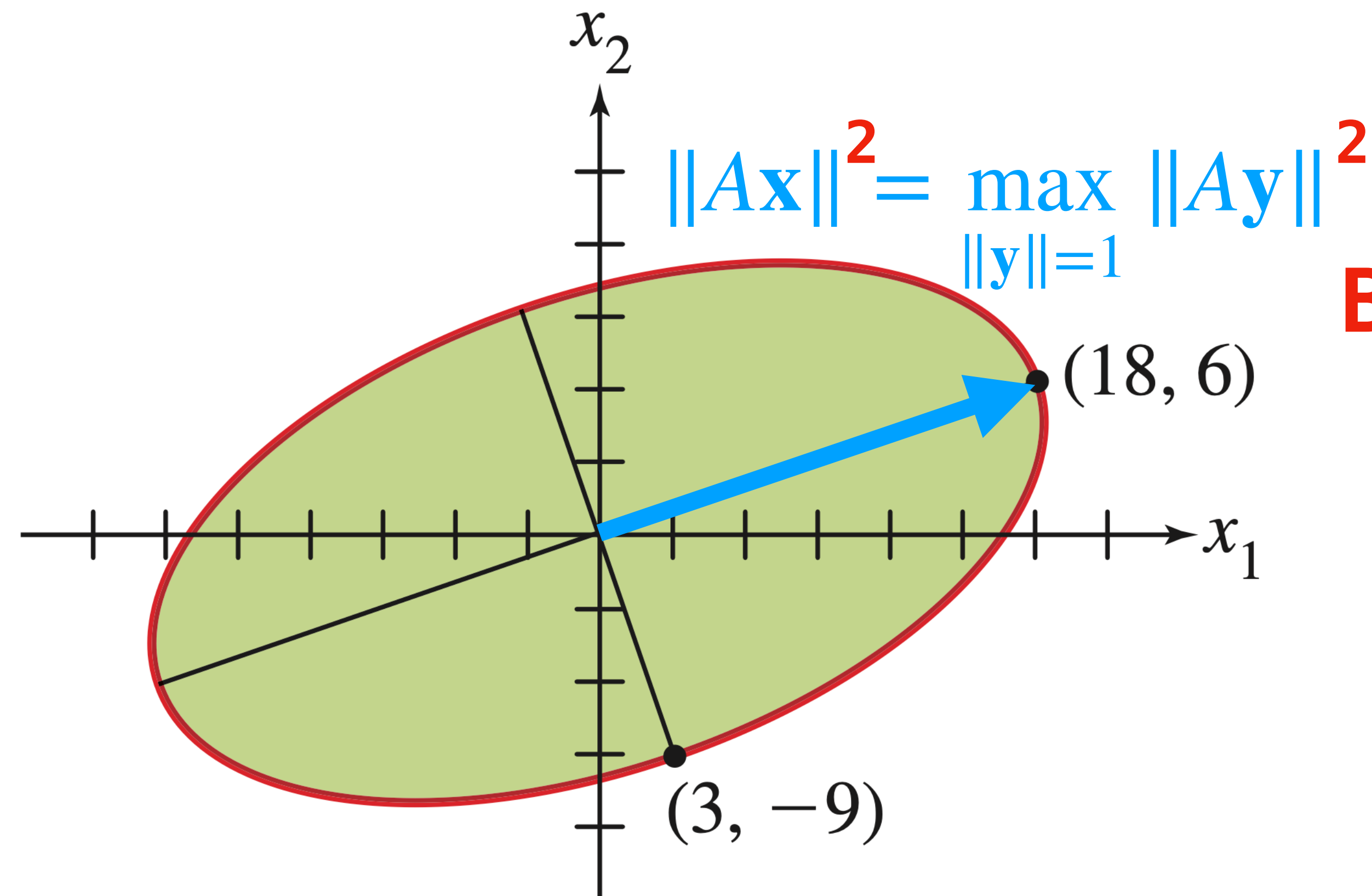


The Picture



This is not a quadratic form...

The Picture



But this is

This is not a quadratic form...

A Quadratic Form

What does $\|A\mathbf{x}\|^2$ look like?

$$\|v\| = \sqrt{\langle v, v \rangle}$$

$$\|A\vec{x}\|^2 = \langle A\vec{x}, A\vec{x} \rangle$$

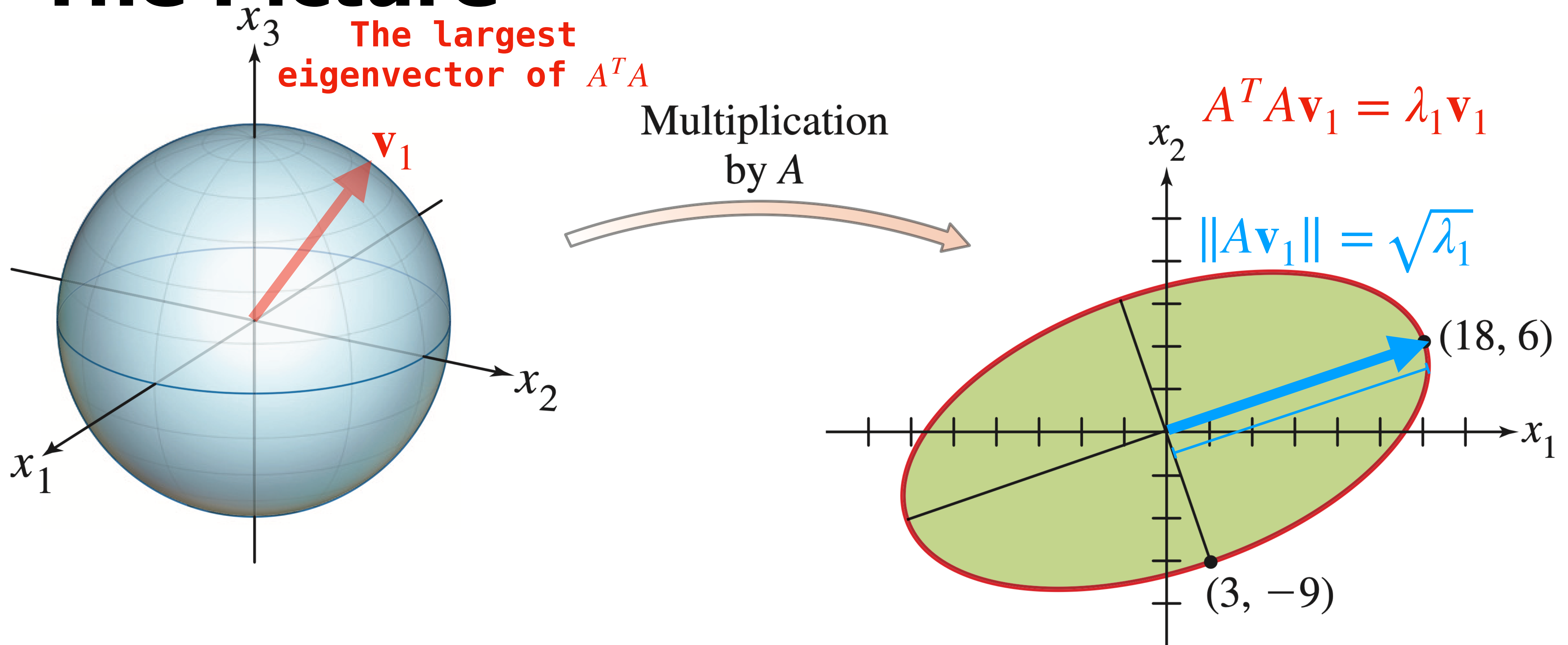
$$= (A\vec{x})^T A\vec{x}$$

$$= \vec{x}^T (A^T A) \vec{x}$$

$$(A^T A)^T = A^T (A^T)^T = \underline{A^T A}$$

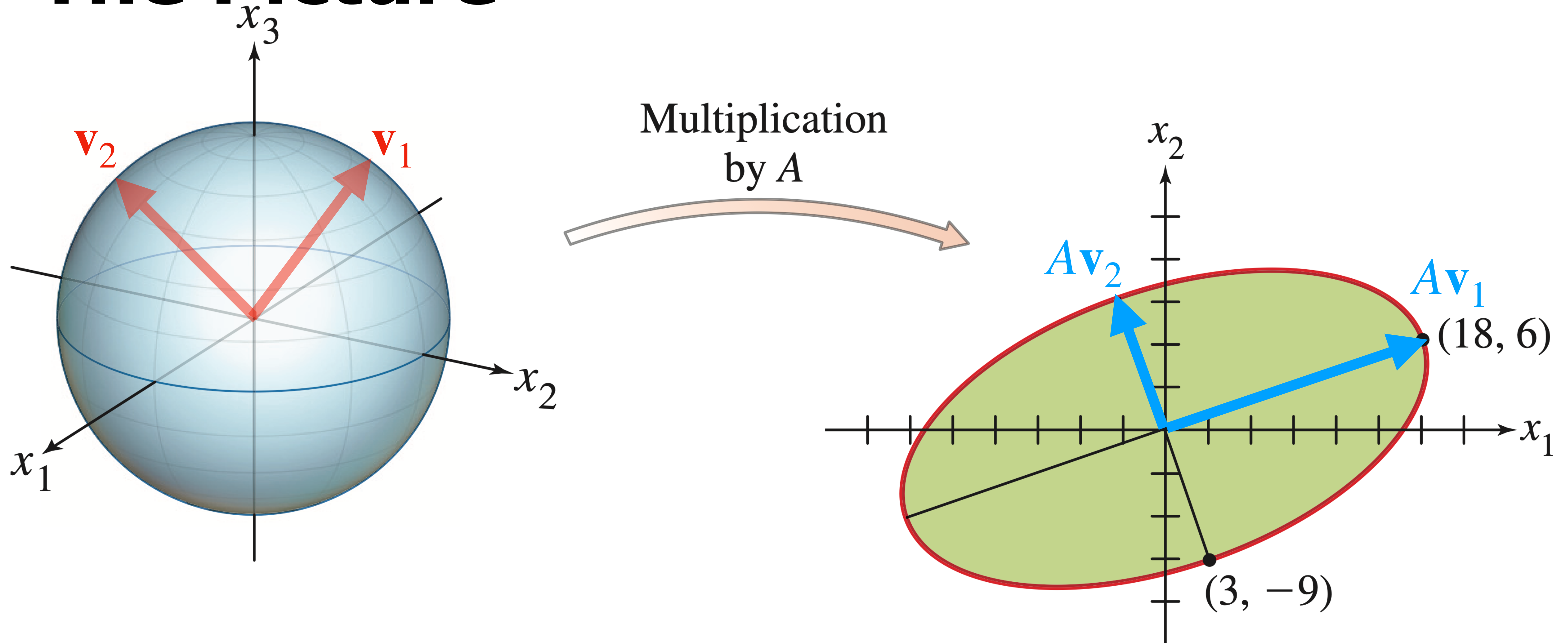
symmetric

The Picture



$\mathbf{v}_1$ solves the constrained optimization problem

The Picture



The second eigenvector is sent to the *minimum* principle axis

Properties of $A^T A$

Properties of $A^T A$

It's symmetric

Properties of $A^T A$

It's symmetric

So its *orthogonally diagonalizable*

Properties of $A^T A$

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So its *orthogonally diagonalizable*

There is an orthogonal basis of eigenvectors

Properties of $A^T A$

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There is an orthogonal basis of eigenvectors

It's eigenvalues are nonnegative

Properties of $A^T A$

It's symmetric

So its *orthogonally diagonalizable*

There is an orthogonal basis of eigenvectors

It's eigenvalues are nonnegative

It's positive semidefinite

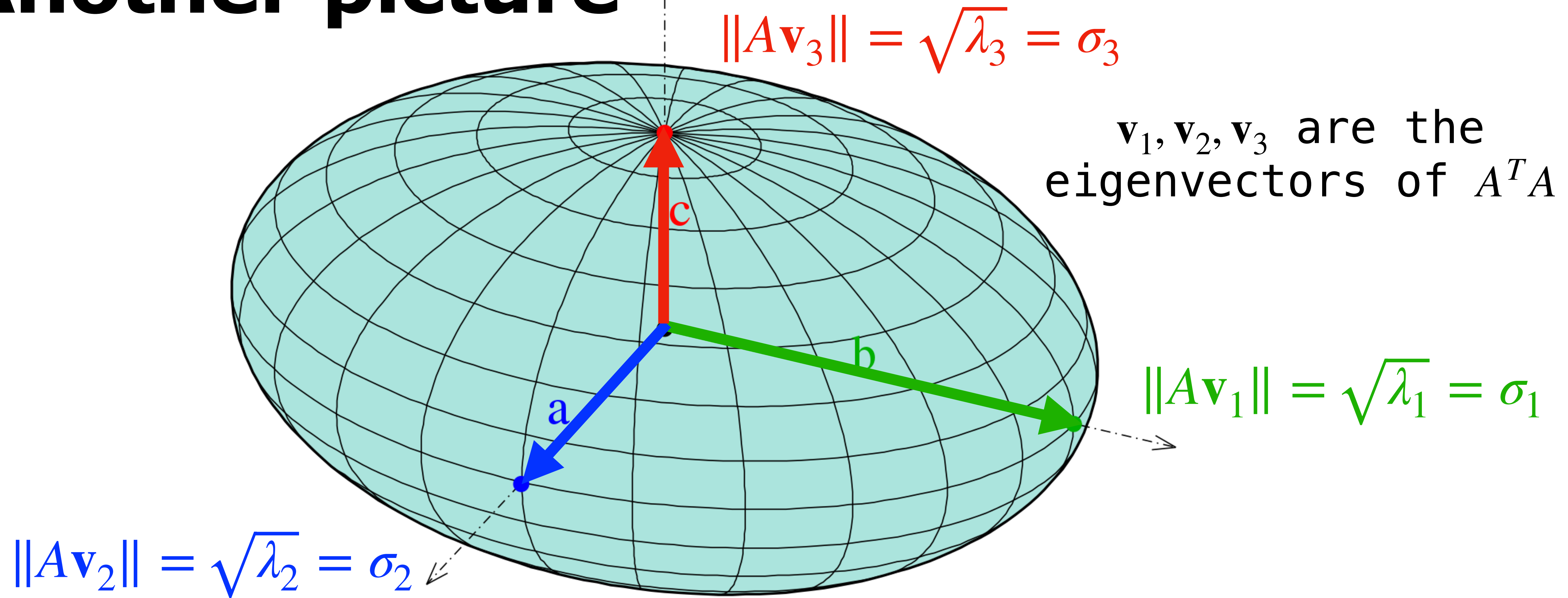
Singular Values

Def. For $A \in \mathbb{R}^{m \times n}$ the **singular values** of A are the n values

$$\sigma_1 \geq \sigma_2 \dots \geq \sigma_n \geq 0$$

where $\sigma_i = \sqrt{\lambda_i}$ and λ_i is an eigenvalue of $A^T A$

Another picture



The **singular values** are the lengths of the *axes of symmetry* of the ellipsoid after transforming the unit sphere

Every $m \times n$ matrix transforms the
unit ~~m~~ _{n} -sphere into an ~~n~~ _{m} -ellipsoid

So every $m \times n$ matrix has n
singular values

What else can we say?

Let $\mathbf{v}_1, \dots, \mathbf{v}_n$ be an **orthogonal** eigenbasis of $\mathbb{R}^n$ for $A^T A$ and suppose A has r *nonzero* singular values

Symmetric

Thm. $A\mathbf{v}_1, \dots, A\mathbf{v}_r$ is an orthogonal basis of $\text{Col}(A)$

What else can we say?

Let $\mathbf{v}_1, \dots, \mathbf{v}_n$ be an **orthogonal** eigenbasis of $\mathbb{R}^n$ for $A^T A$ and suppose A has r *nonzero* singular values

Thm. $A\mathbf{v}_1, \dots, A\mathbf{v}_r$ is an orthogonal basis of $\text{Col}(A)$

This is the most important theorem for SVD

Verifying it

Let's show $\underbrace{A\mathbf{v}_1, \dots, A\mathbf{v}_r}_{}$ are orthogonal (and L.I.)

$$\langle A\mathbf{v}_i, A\mathbf{v}_j \rangle = \quad i \neq j$$

$$(A\mathbf{v}_i)^T (A\mathbf{v}_j) =$$

$$\mathbf{v}_i^T A^T A \mathbf{v}_j = \mathbf{v}_i^T \lambda_j \vec{\mathbf{v}}_j = \lambda_j \mathbf{v}_i^T \mathbf{v}_j$$

$$= \lambda_j \langle \mathbf{v}_i, \mathbf{v}_j \rangle$$

$$= 0$$

Verifying it

$$A = [\vec{a}_1, \dots, \vec{a}_n]$$

Let's show $A\mathbf{v}_1, \dots, A\mathbf{v}_r$ span $\text{Col}(A)$

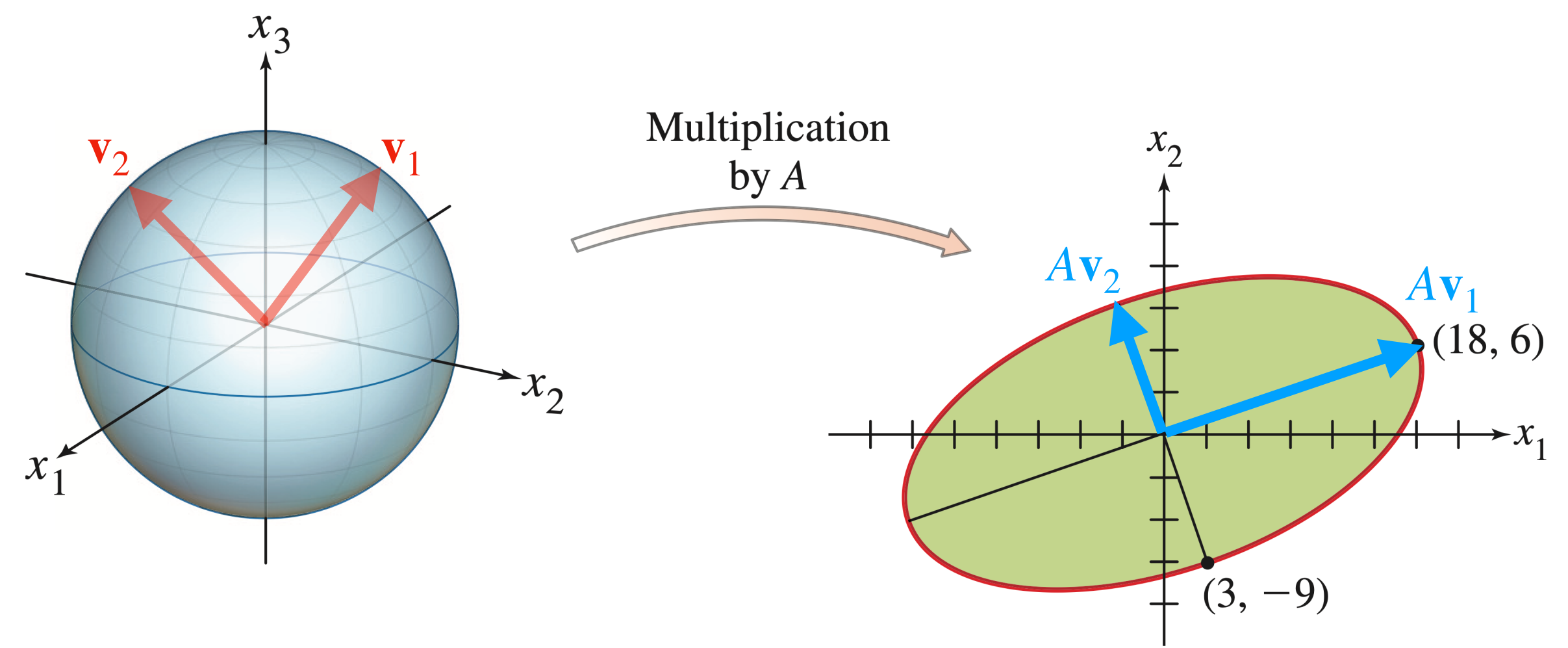
$$\vec{u} \in \text{Col}(A)$$

$$\vec{u} = \sum_i \alpha_i \vec{a}_i = A \underbrace{\vec{w}}_{\mathbb{R}^n}$$

$$= A \left(\sum_{k=1}^r \beta_k \vec{v}_k \right)$$

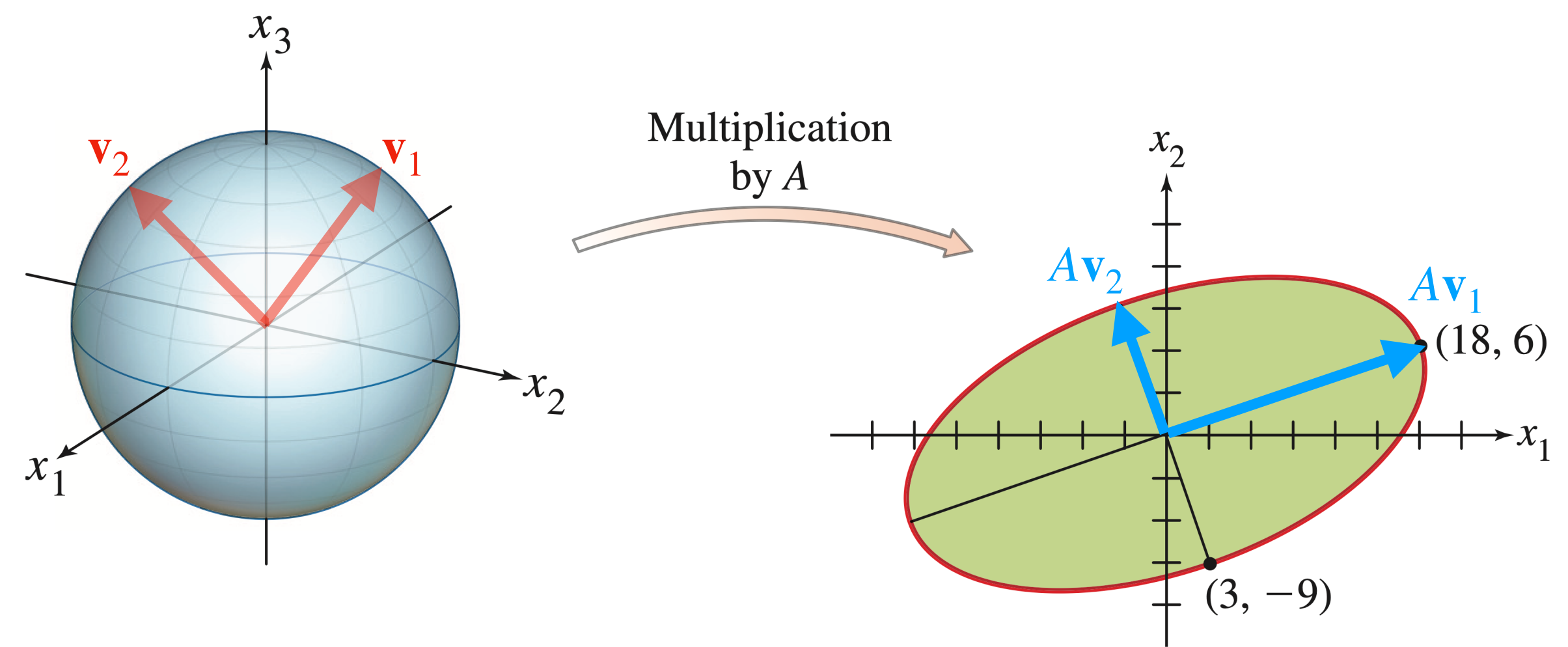
$$\sum_{k=1}^r \beta_k (A \vec{v}_k)$$

Putting it together



Putting it together

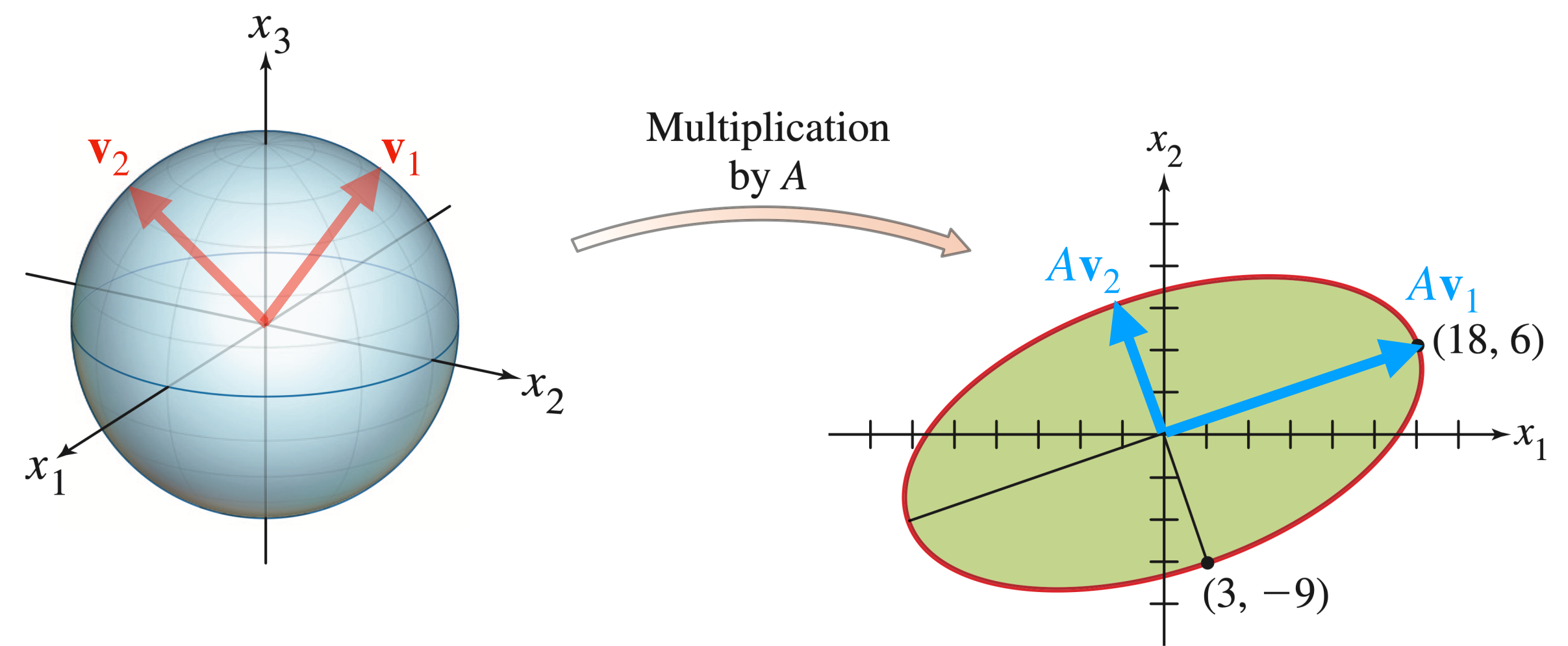
Let $A \in \mathbb{R}^{m \times n}$ be of rank r



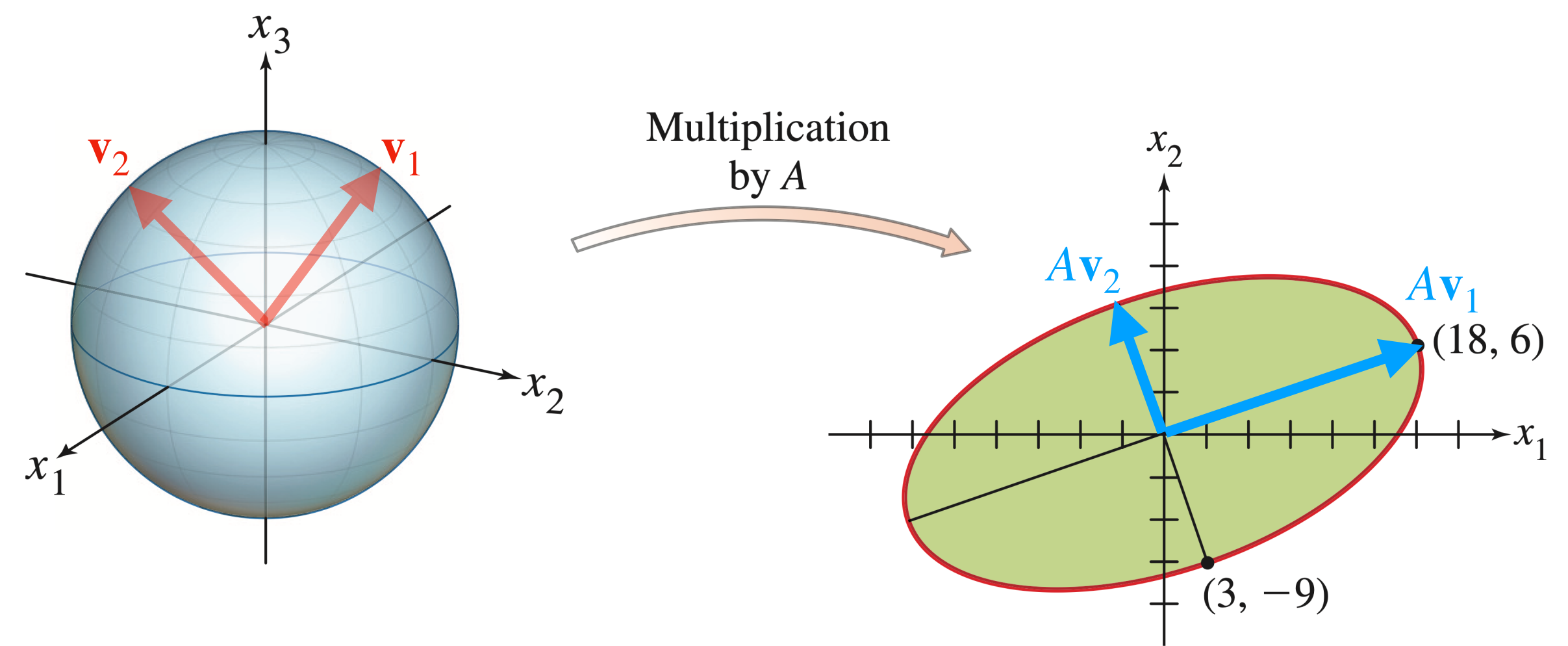
Putting it together

Let $A \in \mathbb{R}^{m \times n}$ be of rank r

What we know:



Putting it together

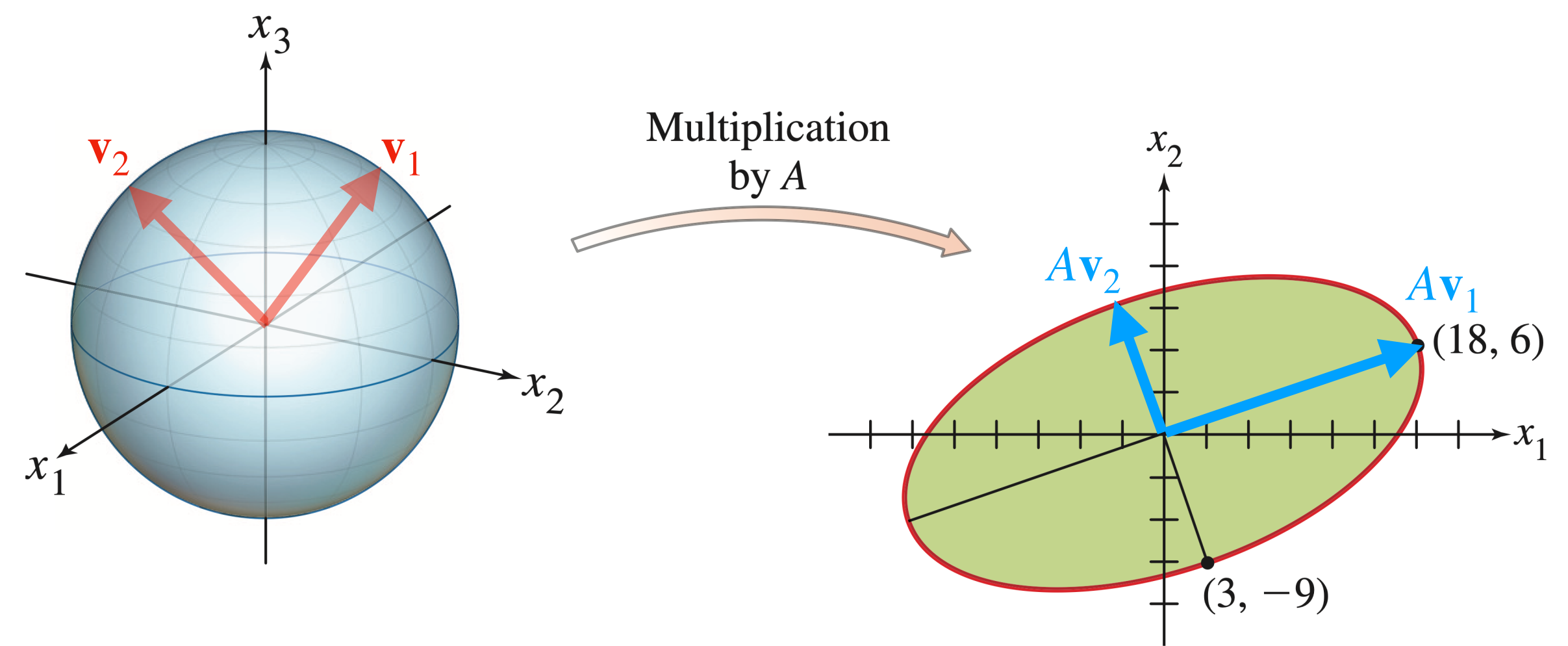


Let $A \in \mathbb{R}^{m \times n}$ be of rank r

What we know:

» We can find orthonormal vectors $\mathbf{v}_1, \dots, \mathbf{v}_n$ in $\mathbb{R}^n$ such that $A\mathbf{v}_1, \dots, A\mathbf{v}_r$ in $\mathbb{R}^m$ form an orthogonal basis for $\text{Col}(A)$

Putting it together



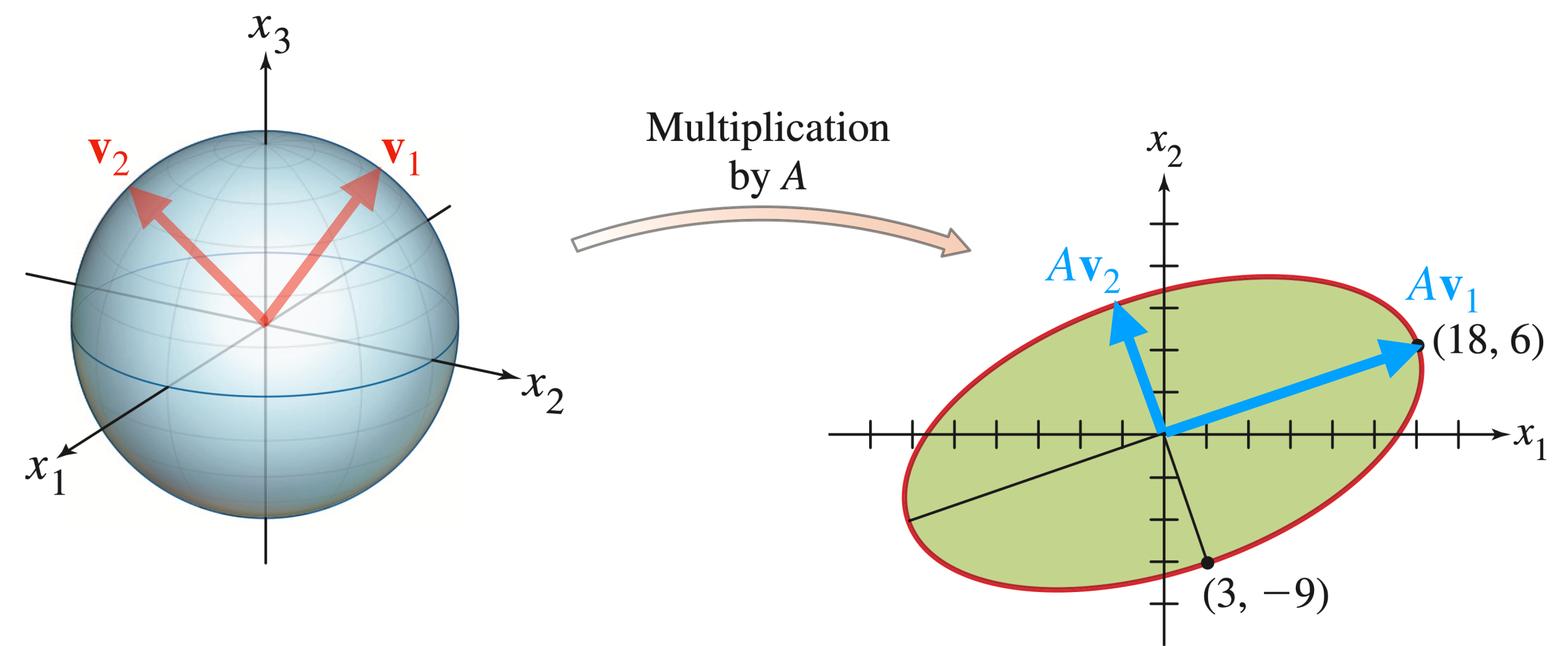
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» So if we take $\mathbf{u}_i = \frac{A\mathbf{v}_i}{\|A\mathbf{v}_i\|}$, we get an **orthonormal** basis of $\text{Col}(A)$

Putting it together



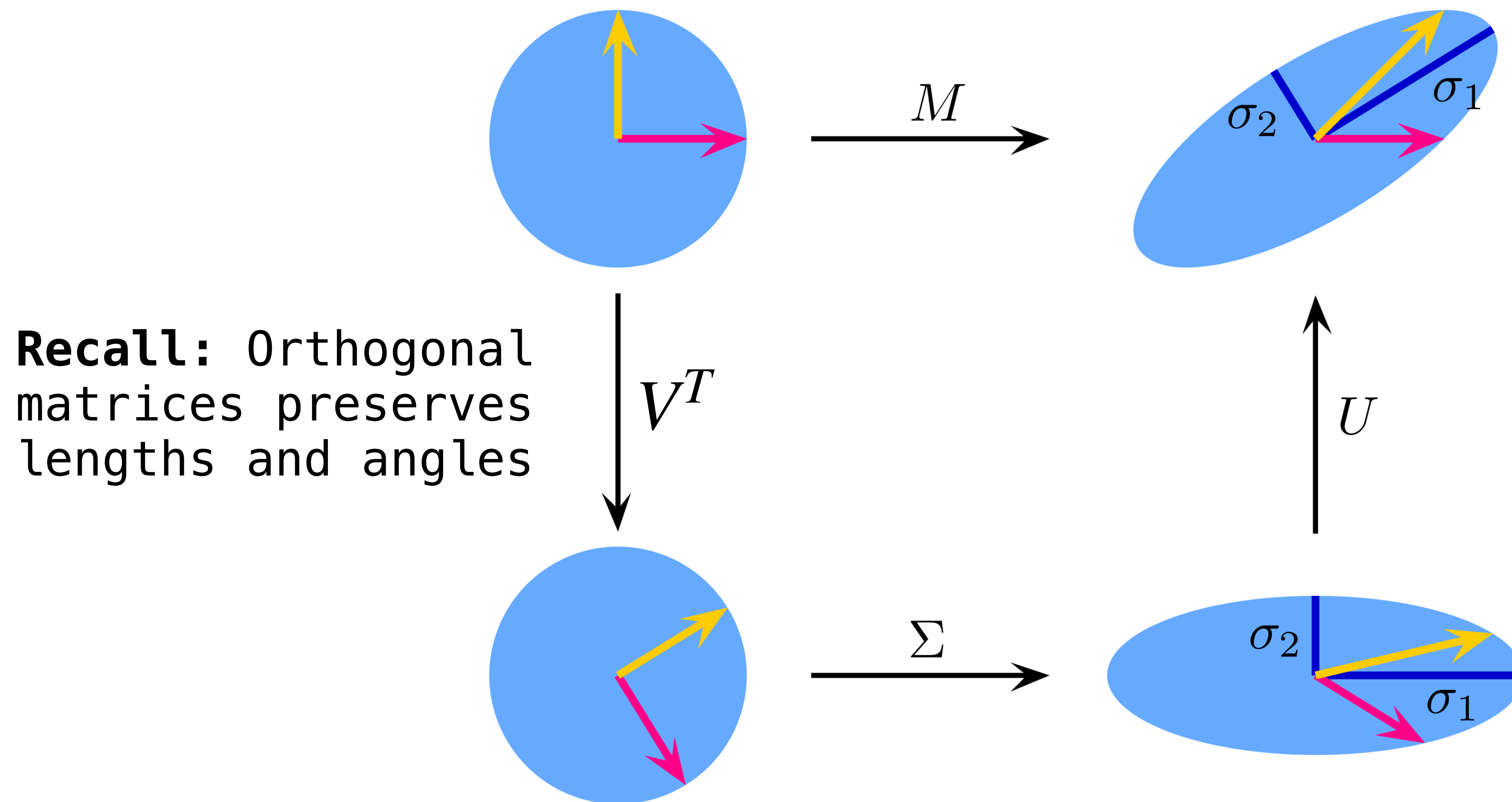
Let $A \in \mathbb{R}^{m \times n}$ be of rank r

What we know:

- » We can find orthonormal vectors $\mathbf{v}_1, \dots, \mathbf{v}_n$ in $\mathbb{R}^n$ such that $A\mathbf{v}_1, \dots, A\mathbf{v}_r$ in $\mathbb{R}^m$ form an orthogonal basis for $\text{Col}(A)$
- » So if we take $\mathbf{u}_i = \frac{A\mathbf{v}_i}{\|A\mathbf{v}_i\|}$, we get an **orthonormal** basis of $\text{Col}(A)$
- » And we can extend this to $\mathbf{u}_1, \dots, \mathbf{u}_m$ an orthonormal basis of $\mathbb{R}^m$ (via Gram–Schmidt)

Singular Value Decomposition

High Level View of the Decomposition



$$M = U \cdot \Sigma \cdot V^T$$

The Important Equality

$$A\mathbf{v}_i = \|A\mathbf{v}_i\|\mathbf{u}_i = \sigma_i\mathbf{u}_i$$

$$\mathbf{u}_i = \frac{A\mathbf{v}_i}{\|A\mathbf{v}_i\|}$$

sign
length

The Important Equality

$$\mathbf{u}_i = \frac{A\mathbf{v}_i}{\|A\mathbf{v}_i\|}$$

$$A\mathbf{v}_i = \|A\mathbf{v}_i\|\mathbf{u}_i = \sigma_i\mathbf{u}_i$$

Remember that $\sigma_i = \sqrt{\lambda_i}$ is the singular value,
which is the length $\|A\mathbf{v}_i\|$

The Important Equality

$$\mathbf{u}_i = \frac{A\mathbf{v}_i}{\|A\mathbf{v}_i\|}$$

$$A\mathbf{v}_i = \|A\mathbf{v}_i\|\mathbf{u}_i = \sigma_i\mathbf{u}_i$$

Remember that $\sigma_i = \sqrt{\lambda_i}$ is the singular value,
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What happens when we write this in matrix form?

The Important Equality

$$A[\mathbf{v}_1 \ \dots \ \mathbf{v}_n] = [\sigma_1 \mathbf{u}_1 \ \dots \ \sigma_n \mathbf{u}_n]$$

Remember that $\sigma_i = \sqrt{\lambda_i}$ is the singular value,
which is the length $\|A\mathbf{v}_i\|$

The Important Equality

$$A[\mathbf{v}_1 \ \dots \ \mathbf{v}_n] = [\sigma_1 \mathbf{u}_1 \ \dots \ \sigma_n \mathbf{u}_n]$$

Remember that $\sigma_i = \sqrt{\lambda_i}$ is the singular value, which is the length $\|A\mathbf{v}_i\|$

Let's take $V = [\mathbf{v}_1 \ \dots \ \mathbf{v}_n]$ and $U = [\mathbf{u}_1 \ \dots \ \mathbf{u}_m]$ and

The Important Equality

$$A[\mathbf{v}_1 \ \dots \ \mathbf{v}_n] = [\sigma_1 \mathbf{u}_1 \ \dots \ \sigma_n \mathbf{u}_n]$$

Remember that $\sigma_i = \sqrt{\lambda_i}$ is the singular value, which is the length $\|A\mathbf{v}_i\|$

Let's take $V = [\mathbf{v}_1 \ \dots \ \mathbf{v}_n]$ and $U = [\mathbf{u}_1 \ \dots \ \mathbf{u}_m]$ and

eigenvectors of $A^T A$

$$\mathbf{u}_i = \frac{A\mathbf{v}_i}{\|A\mathbf{v}_i\|}$$

$$\Sigma = \begin{matrix} m > n \\ \begin{bmatrix} \sigma_1 & \dots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \dots & \sigma_n \\ 0 & \dots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \dots & 0 \end{bmatrix} \end{matrix} \text{ or } \Sigma = \begin{matrix} m < n \\ \begin{bmatrix} \sigma_1 & \dots & 0 & 0 & \dots & 0 \\ \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ 0 & \dots & \sigma_m & 0 & \dots & 0 \end{bmatrix} \end{matrix} \text{ or } \Sigma = \begin{matrix} m = n \\ \begin{bmatrix} \sigma_1 & \dots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \dots & \sigma_n \end{bmatrix} \end{matrix}$$

The Important Equality

$$A[\mathbf{v}_1 \ \dots \ \mathbf{v}_n] = [\sigma_1 \mathbf{u}_1 \ \dots \ \sigma_n \mathbf{u}_n]$$

Remember that $\sigma_i = \sqrt{\lambda_i}$ is the singular value, which is the length $\|A\mathbf{v}_i\|$

Let's take $V = [\mathbf{v}_1 \ \dots \ \mathbf{v}_n]$ and $U = [\mathbf{u}_1 \ \dots \ \mathbf{u}_m]$ and

remember: U is orthonormal

$$\Sigma = \begin{matrix} m > n \\ \begin{bmatrix} \sigma_1 & \dots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \dots & \sigma_n \\ 0 & \dots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \dots & 0 \end{bmatrix} \end{matrix} \text{ or } \Sigma = \begin{matrix} m < n \\ \begin{bmatrix} \sigma_1 & \dots & 0 & 0 & \dots & 0 \\ \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ 0 & \dots & \sigma_m & 0 & \dots & 0 \end{bmatrix} \end{matrix} \text{ or } \Sigma = \begin{matrix} m = n \\ \begin{bmatrix} \sigma_1 & \dots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \dots & \sigma_n \end{bmatrix} \end{matrix}$$

The Important Equality

$$AV = U\Sigma$$

Remember that $\sigma_i = \sqrt{\lambda_i}$ is the singular value, which is the length $\|A\mathbf{v}_i\|$

Let's take $V = [\mathbf{v}_1 \dots \mathbf{v}_n]$ and $U = [\mathbf{u}_1 \dots \mathbf{u}_m]$ and

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The Important Equality

$$\overset{m \times n}{A} \underset{n \times n}{V} = \overset{m \times m}{U} \underset{m \times n}{\Sigma}$$

Remember that $\sigma_i = \sqrt{\lambda_i}$ is the singular value, which is the length $\|A\mathbf{v}_i\|$

Let's take $V = [\mathbf{v}_1 \dots \mathbf{v}_n]$ and $U = [\mathbf{u}_1 \dots \mathbf{u}_m]$ and

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$$\Sigma = \overset{m > n}{\begin{bmatrix} \sigma_1 & \dots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \dots & \sigma_n \\ 0 & \dots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \dots & 0 \end{bmatrix}} \quad \text{or} \quad \Sigma = \overset{m < n}{\begin{bmatrix} \sigma_1 & \dots & 0 & 0 & \dots & 0 \\ \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ 0 & \dots & \sigma_m & 0 & \dots & 0 \end{bmatrix}} \quad \text{or} \quad \Sigma = \overset{m = n}{\begin{bmatrix} \sigma_1 & \dots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \dots & \sigma_n \end{bmatrix}}$$

The Important Equality

if $A \in \mathbb{R}^{n \times n}$ is orthonormal
then $A^{-1} = A^T$

$$AVV^T = U\Sigma V^T$$

Remember that $\sigma_i = \sqrt{\lambda_i}$ is the singular value,
which is the length $\|A\mathbf{v}_i\|$

Let's take $V = [\mathbf{v}_1 \dots \mathbf{v}_n]$ and $U = [\mathbf{u}_1 \dots \mathbf{u}_m]$ and

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The Important Equality

$$A = U\Sigma V^T$$

Remember that $\sigma_i = \sqrt{\lambda_i}$ is the singular value, which is the length $\|A\mathbf{v}_i\|$

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The Important Equality

singular value decomposition

$$A = U \Sigma V^T$$

Remember that $\sigma_i = \sqrt{\lambda_i}$ is the singular value, which is the length $\|A\mathbf{v}_i\|$

Let's take $V = [\mathbf{v}_1 \dots \mathbf{v}_n]$ and $U = [\mathbf{u}_1 \dots \mathbf{u}_m]$ and

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$$\Sigma = \begin{matrix} m > n \\ \begin{bmatrix} \sigma_1 & \dots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \dots & \sigma_n \\ 0 & \dots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \dots & 0 \end{bmatrix} \end{matrix} \text{ or } \Sigma = \begin{matrix} m < n \\ \begin{bmatrix} \sigma_1 & \dots & 0 & 0 & \dots & 0 \\ \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ 0 & \dots & \sigma_m & 0 & \dots & 0 \end{bmatrix} \end{matrix} \text{ or } \Sigma = \begin{matrix} m = n \\ \begin{bmatrix} \sigma_1 & \dots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \dots & \sigma_n \end{bmatrix} \end{matrix}$$

Singular Value Decomposition

Thm. For $A \in \mathbb{R}^{m \times n}$, there are *orthogonal* matrices $U \in \mathbb{R}^{m \times m}$ and $V \in \mathbb{R}^{n \times n}$ such that

$$A = \overset{m \times m}{\boxed{U}} \underset{m \times n}{\Sigma} \overset{n \times n}{\boxed{V^T}}$$

where diagonal entries* of Σ are $\sigma_1, \dots, \sigma_n$ the singular values of A

* these are diagonal entries in a *non-square* matrix

Singular Value Decomposition

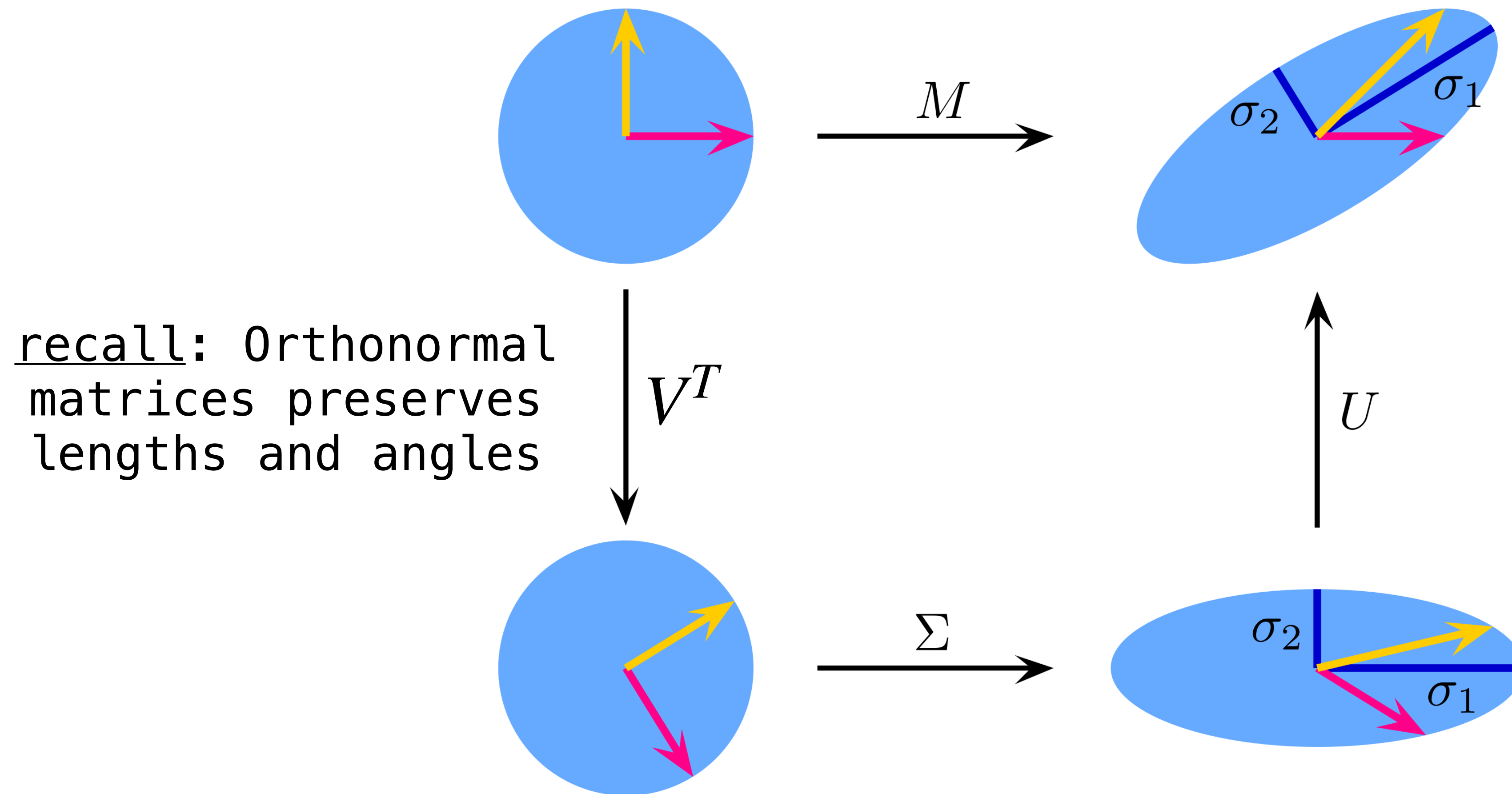
Thm. For $A \in \mathbb{R}^{m \times n}$, there are *orthogonal* matrices
 $U \in \mathbb{R}^{m \times m}$ and $V \in \mathbb{R}^{n \times n}$ such that
left singular *right singular*

$$A = \overset{m \times m}{U} \underset{m \times n}{\Sigma} \overset{n \times n}{V^T}$$

where diagonal entries* of Σ are $\sigma_1, \dots, \sigma_n$ the singular values of A

* these are diagonal entries in a *non-square* matrix

The Picture (Again)



$$M = U \cdot \Sigma \cdot V^T$$

Finding a SVD

Step 1: Set up Σ

$$\begin{bmatrix} 1 & -1 \\ -2 & 2 \\ 2 & -2 \end{bmatrix}$$

The **singular values** are the square roots of the eigenvalues of $A^T A$ (or AA^T)

$$A^T A = \begin{bmatrix} 1 & -2 & 2 \\ -1 & 2 & -2 \\ 2 & -2 & -2 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ -2 & 2 \\ 2 & -2 \end{bmatrix} = \begin{bmatrix} 9 & -9 \\ -9 & 9 \end{bmatrix}$$

$$\det(A^T A - \lambda I) = (9 - \lambda)^2 - 81$$
$$= \lambda^2 - 18\lambda$$

$$\lambda = 18, 0$$

$$A = U \Sigma V^T$$

$$A^T = (U \Sigma V^T)^T$$

$$= \underbrace{V \Sigma^T U^T}_{\text{SVD for } A^T}$$

$$\sigma_1 = 3\sqrt{2}$$

$$\sigma_2 = 0$$

Step 2: Set up V

$$A^T A = \begin{bmatrix} 9 & -9 \\ -9 & 9 \end{bmatrix}$$

$$\sigma_1 = 3\sqrt{2} \quad \sigma_2 = 0$$

$$\begin{bmatrix} 1 & -1 \\ -2 & 2 \\ 2 & -2 \end{bmatrix}$$

Find an orthonormal eigenbasis for $A^T A$

$$A^T A - 18I \sim \begin{bmatrix} -9 & -9 \\ -9 & -9 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix} \quad \begin{array}{l} x_1 = -x_2 \\ x_2 \text{ is free} \end{array}$$

$$\frac{1}{\sqrt{2}} \begin{bmatrix} -1 \\ 1 \end{bmatrix} = \begin{bmatrix} -\sqrt{2}/2 \\ \sqrt{2}/2 \end{bmatrix}$$

$$A^T A - 0I \sim \begin{bmatrix} 9 & -9 \\ -9 & 9 \end{bmatrix} \sim \begin{bmatrix} 1 & -1 \\ 0 & 0 \end{bmatrix} \quad \begin{array}{l} x_1 = x_2 \\ x_2 \text{ is free} \end{array}$$

$$\begin{bmatrix} \sqrt{2}/2 \\ \sqrt{2}/2 \end{bmatrix}$$

$$V = \begin{bmatrix} -\sqrt{2}/2 & \sqrt{2}/2 \\ \sqrt{2}/2 & \sqrt{2}/2 \end{bmatrix}$$

Step 3: Set up U (Part 1)

$$\begin{bmatrix} 1 & -1 \\ -2 & 2 \\ 2 & -2 \end{bmatrix}$$

If $\mathbf{v}_1, \dots, \mathbf{v}_n$ is an eigenbasis of $\mathbb{R}^n$ (in decreasing order of eigenvalue), then $A\mathbf{v}_1, \dots, A\mathbf{v}_r$ is an eigenbasis of $\text{Col}(A)$ (where r is the rank of A). These vectors can be normalized and made the first r columns of U

$$\vec{u}_i = \frac{A \mathbf{v}_i}{\|A \mathbf{v}_i\|} = \frac{A \vec{r}_i}{\sigma_i}$$

$$A \mathbf{v}_1 = A \begin{bmatrix} -1/2 \\ 1 \\ 1/2 \end{bmatrix} \sim A \begin{bmatrix} -1 \\ 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & -1 \\ -2 & 2 \\ 2 & -2 \end{bmatrix} \begin{bmatrix} -1 \\ 2 \end{bmatrix} = \begin{bmatrix} -2 \\ 4 \\ -4 \end{bmatrix}$$

$$u_1 = \frac{1}{\left\| \begin{bmatrix} -2 \\ 4 \\ -4 \end{bmatrix} \right\|} \begin{bmatrix} -2 \\ 4 \\ -4 \end{bmatrix} = \begin{bmatrix} -1/3 \\ 2/3 \\ -2/3 \end{bmatrix}$$

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$$A\vec{v}_i = \sigma_i \vec{u}_i$$

$$A\vec{v}_2 \sim A \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

if $\sigma_i = 0$ then

$$\vec{v}_i \in \text{Nul } A$$

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$$U = \begin{bmatrix} -1/\sqrt{3} & 0 & -4/\sqrt{18} \\ 1/\sqrt{3} & \sqrt{2}/2 & -1/\sqrt{18} \\ -2/\sqrt{3} & \sqrt{2}/2 & 1/\sqrt{18} \end{bmatrix}$$

Step 4: Set up U (Part 2)

$$\begin{bmatrix} 1 & -1 \\ -2 & 2 \\ 2 & -2 \end{bmatrix}$$

If $m > r$, then extend $\mathbf{u}_1, \dots, \mathbf{u}_r$ until it has m orthonormal vectors

$$\mathbf{u}_1 = \begin{bmatrix} -1/3 \\ 2/3 \\ -2/3 \end{bmatrix} \quad \mathbf{u}_2 = \begin{bmatrix} 0 \\ r_2/2 \\ r_2/2 \end{bmatrix}$$

$$\begin{bmatrix} -1 & 2 & -2 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} -1 & 2 & -2 \\ 0 & 1 & 1 \end{bmatrix} \sim \begin{bmatrix} 1 & -2 & 2 \\ 0 & 1 & 1 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 4 \\ 0 & 1 & 1 \end{bmatrix}$$

$x_1 = -4x_3$
 $x_2 = -x_3$
 x_3 is free

$$\begin{bmatrix} -4 \\ -1 \\ 1 \end{bmatrix} \sim \begin{bmatrix} -4/\sqrt{18} \\ -1/\sqrt{18} \\ 1/\sqrt{18} \end{bmatrix}$$

$\mathbf{u}_3$

Step 5: Put it all together

$$A = U \Sigma V^T$$

$$\begin{bmatrix} 1 & -1 \\ -2 & 2 \\ 2 & -2 \end{bmatrix}$$

$$A = \underbrace{\begin{bmatrix} -1/3 & 0 & -4/\sqrt{10} \\ 2/3 & \sqrt{2}/2 & -1/\sqrt{2} \\ -2/3 & \sqrt{2}/2 & 1/\sqrt{10} \end{bmatrix}}_U \underbrace{\begin{bmatrix} \overset{\sigma_1}{3\sqrt{2}} & 0 & 0 \\ 0 & \overset{\sigma_2}{\sqrt{2}} & 0 \\ 0 & 0 & 0 \end{bmatrix}}_{\Sigma} \underbrace{\begin{bmatrix} -\sqrt{2}/2 & \sqrt{2}/2 \\ \sqrt{2}/2 & \sqrt{2}/2 \end{bmatrix}}_{V^T}$$

SVD in NumPy

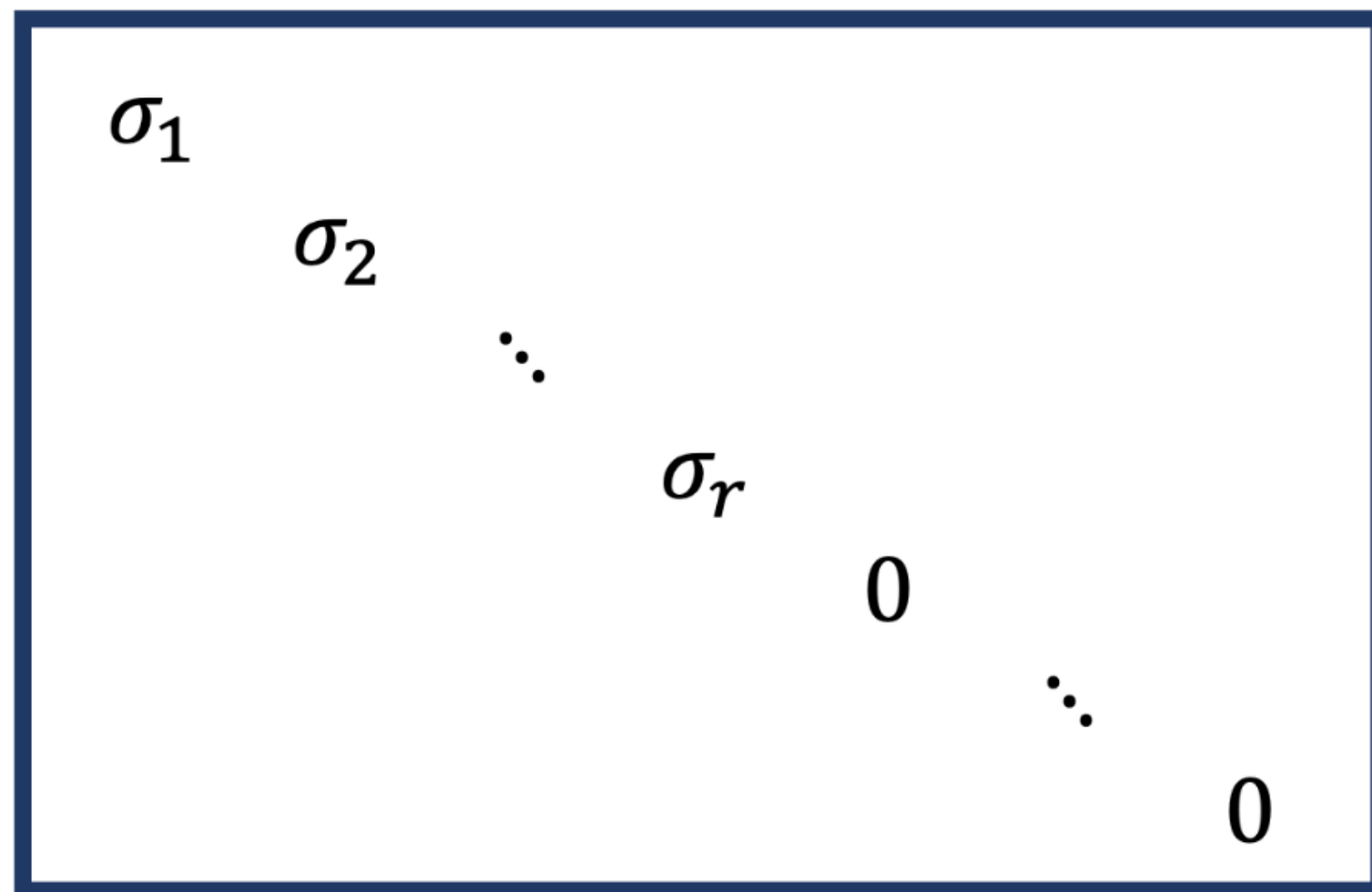
In reality, we will almost never build SVDs by hand. We can use:

`numpy.linalg.svd`

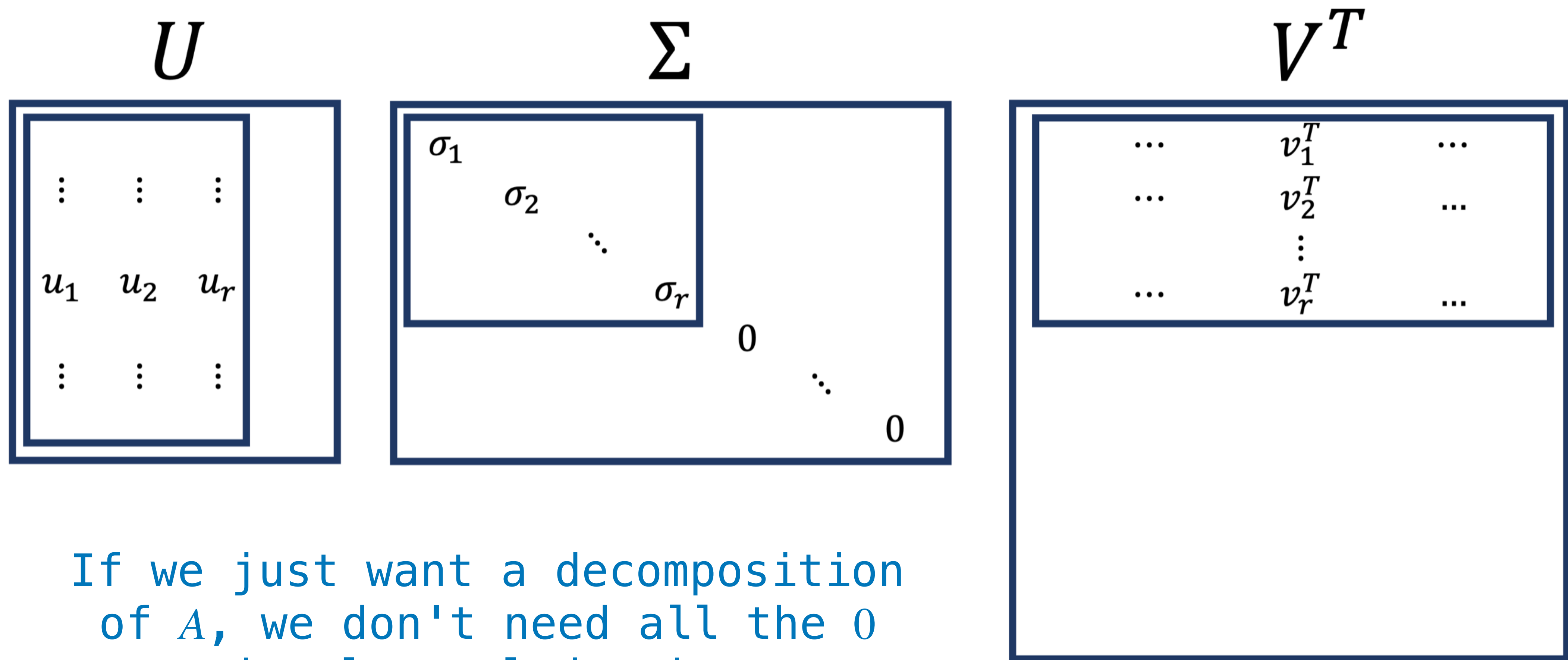
Let's do a quick demo...

Pseudoinverses

SVD (The Picture)

 U  Σ  V^T 

Reduced SVD (The Picture)



The Reduced SVD

Thm. For every matrix A of rank r , there is an orthonormal matrix $U \in \mathbb{R}^{m \times r}$, a diagonal matrix $\Sigma \in \mathbb{R}^{r \times r}$ with **positive** entries on the diagonal, and an orthonormal matrix $V \in \mathbb{R}^{n \times r}$ such that

$$A = U\Sigma V^T$$

Back to the Example

$$\begin{bmatrix} 1 & -1 \\ -2 & 2 \\ 2 & -2 \end{bmatrix}$$

$$A = \overset{3 \times 1}{U} \overset{1 \times 1}{\Sigma} \overset{1 \times 3}{V^T}$$
$$\begin{bmatrix} -1/3 \\ 2/3 \\ -2/3 \end{bmatrix} \quad \begin{bmatrix} 3\sigma_2 \end{bmatrix} \quad \begin{bmatrix} -\sigma_2/2 & \sigma_2/2 \end{bmatrix}$$

The Pseudoinverse

Def. Given a reduced SVD $A = U\Sigma V^T$, the **pseudoinverse** of A is $A^+ = V\Sigma^{-1}U^T$

Thm. $A^+\mathbf{b}$ is the *minimum length* **least squares solution** of $A\mathbf{x} = \mathbf{b}$

(in Python we have *`numpy.linalg.pinv`*)

Recall: Least Squares in NumPy

numpy.linalg.lstsq

`linalg.lstsq(a, b, rcond='warn')`

[\[source\]](#)

Return the least-squares solution to a linear matrix equation.

Computes the vector x that approximately solves the equation $a @ x = b$. The equation may be under-, well-, or over-determined (i.e., the number of linearly independent rows of a can be less than, equal to, or greater than its number of linearly independent columns). If a is square and of full rank, then x (but for round-off error) is the “exact” solution of the equation. Else, x minimizes the Euclidean 2-norm $||b - ax||$. If there are multiple minimizing solutions, the one with the smallest 2-norm $||x||$ is returned.

Parameters: a : (M, N) *array_like*

“Coefficient” matrix.

b : $\{(M,), (M, K)\}$ *array_like*

Ordinate or “dependent variable” values. If b is two-dimensional, the least-squares solution is calculated for each of the K columns of b .

$rcond$: *float. optional*

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(why?...))

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(why?...))

because they use SVD!